

Particulate Matter

Week in Review

Week ending 29 August 2026 | 23 August 2026 to 29 August 2026 | 109 records pooled from 7 daily issues

109

RECORDS POOLED
(107 UNIQUE DOIs)

41%

ON THE MEASURE-
MENT
SIDE (45 RECORDS)

10

SUBTOPICS
ACTIVE

37

POOLED EFFECT
ESTIMATES

8

ABSTRACT-FREE
(7% OF WEEK)

The week in one paragraph

The health side took the week back from the measurement side, and the abstract-free share collapsed from 20% to 7% — but the week's real content is a convergence: three independent lines all say the health-relevant quantity is a *composition*, and none of them is measurable by the instruments the epidemiology relies on.

The composition thread runs the week end to end. Li et al. (28 Aug, *Environ Sci Technol Lett*) decompose US county AMI mortality into $\text{PM}_{2.5}$ constituents and find SO_4^{2-} carrying weight 0.762 of a mixture effect (RR 1.116 per quartile) that is several times the total-mass estimate (RR 1.019 per IQR), with the dominant constituent flipping east-to-west from sulfate to organic carbon. Wang et al. (28 Aug, *Sleep Med*) run the same five-constituent decomposition through 19,505 CHARLS participants and land on the secondary inorganic ions again. Zhang et al. (28 Aug, *ACP*) supply the physics: single-particle SP2 and SPAMS measurement in Shenzhen shows nighttime heterogeneous and daytime photochemical aging arrive at nearly the same **mass absorption cross-section enhancement** — ~ 0.8 and $\sim 1.0 \text{ m}^2 \text{ g}^{-1}$ at 532 nm — by chemically different routes. A mass-only optical monitor re-expresses that diurnal coating cycle as concentration drift.

The counter-observation is that the coarse fraction refuses to go away. Park et al. (29 Aug) put PM_{10} ahead of $\text{PM}_{2.5}$ for the high-TSH/low-FT4 thyroid phenotype (aOR 1.55 vs. 1.25 per IQR), and Cliff et al. (29 Aug) identify atmospheric mineral dust as the strongest single predictor of meningococcal outbreaks across the African meningitis belt. Neither is causally identified — both rest on collinear exposure fields — but the fine-fraction default is doing less work in this week's data than the literature's framing assumes.

Operationally the week improved. Abstract-free deposits fell from 22 of 112 (20%) to 8 of 109 (7%), so *Metadata only* is no longer the largest architecture label; *Measurement campaign* (24) is. Two defects offset that: two records were published twice under different daily issues, and five short-label collisions were created and repaired inside this period. Both are recorded below.

Subtopic trend and inflection

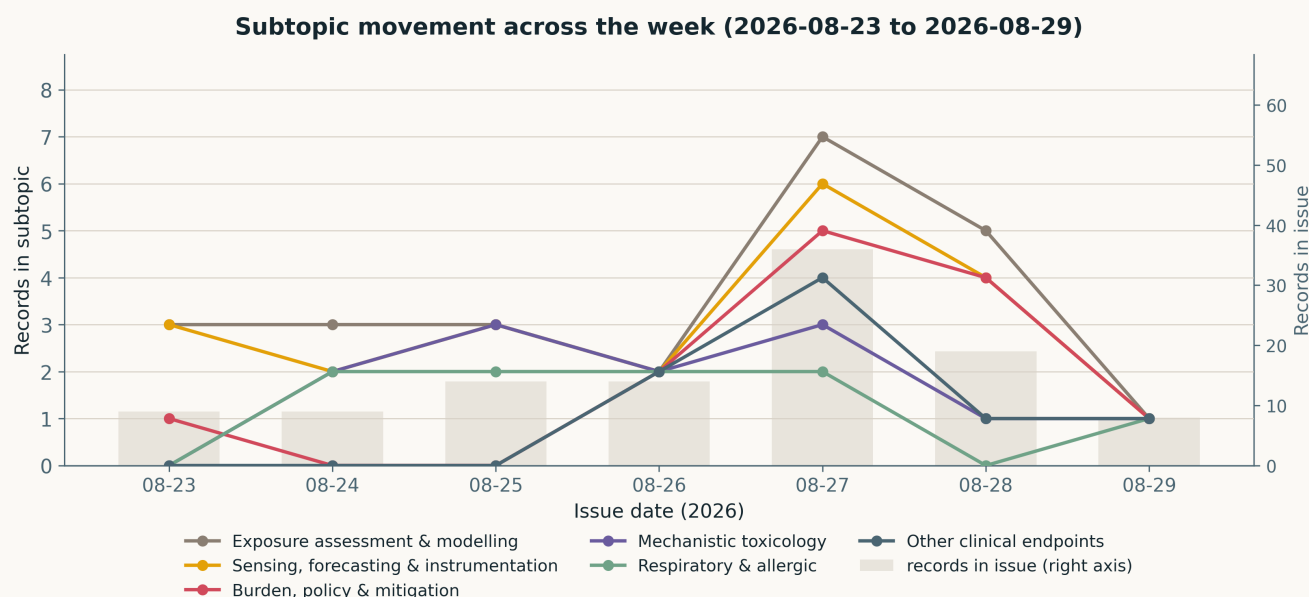


Fig. Per-issue subtopic counts across the week from `state/metrics.csv`. **Read the denominators first:** daily totals were 9, 9, 14, 14, 36, 19 and 8 — a 4.5× range, far more volatile than last week's 2.2×. The 27 August figure of 36 is the largest single-day issue this watch has produced outside the founding backfill, and it is a publisher artefact: a single MDPI *Toxics* volume release contributed six records into a 43-deposit Crossref day. **The trend line for 27 August therefore cannot be read at face value**, and every week-over-week movement below should be discounted accordingly.

The measurement side gave back last week's gain. *Exposure assessment & modelling* fell 36 → 24 while *Sensing, forecasting & instrumentation* held at 22 → 21; together they are 45 of 109, a measurement-side share of 41%, against 52%, 45% and 54% in the three preceding weeks. The four-week sequence 54, 45, 52, 41 has no trend in it — it is the range this pipeline's own harvesting legs produce as the Crossref by-ISSN sweep competes with PubMed for share of a given day. No claim about the field should rest on it.

Expanding. *Burden, policy & mitigation* rose 9 → 13 and *Mechanistic toxicology* 8 → 12, the latter reversing last week's break in a two-week run at 11. *Neuro / mental health* (4 → 7), *Occupational & indoor* (4 → 7) and *Cardiovascular & metabolic* (3 → 5) all recovered from lows that the 22 August rollup flagged as worth a second week of watching; cardiovascular at 5 is still below its running level and still well under the respiratory count.

Contracting. *Other clinical endpoints* fell 13 → 8, partially unwinding the two-week finding that the clinical tail had outgrown the cardiorespiratory trunk. *Reproductive & developmental* at 3 is the week's smallest cluster and its lowest reading this month — notable against last week, when the developmental mixture cohorts supplied most of the constituent-resolved estimates. **All ten subtopics were active on at least one day, for the fifth consecutive week.**

Study architecture

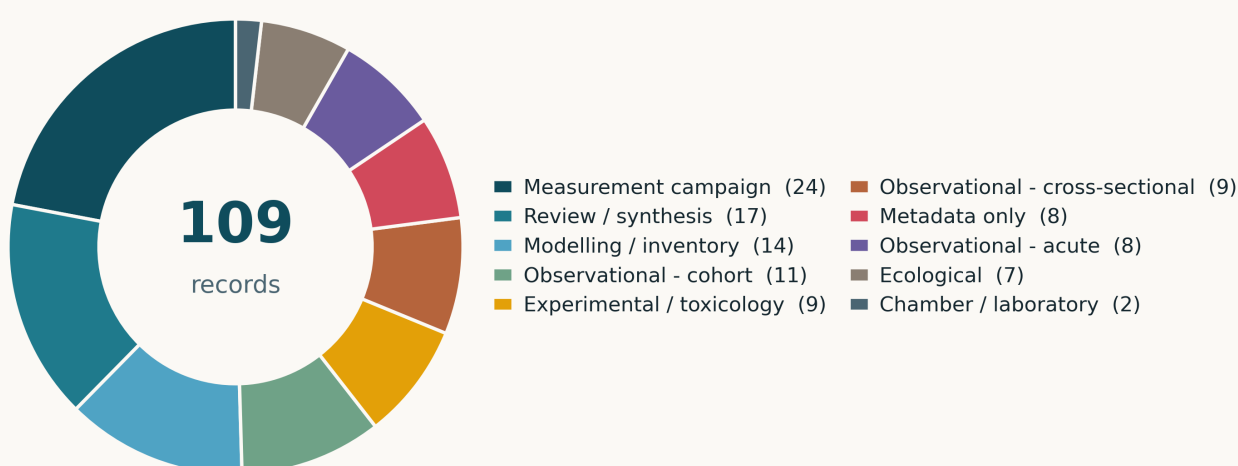


Fig. Pooled study architecture over 109 records. **Metadata only** has fallen from 22 records (20% of last week) to 8 (7%) and is no longer the largest label — the title-pattern front-matter filter added on 22 August, plus a deliberate decision across this week's issues to reject abstract-free deposits to a retry list rather than carry them at tier C, both contributed. *Measurement campaign* (24) now leads, ahead of *Review / synthesis* (17) and *Modelling / inventory* (14); *Observational — cohort* (11) and *Experimental / toxicology* (9) are the load-bearing evidence classes. ***Trial / intervention* is 0 records in 109** — for the fifth week running no completed randomised PM intervention entered the corpus, and this week not even a process evaluation did.

Pooled analytics

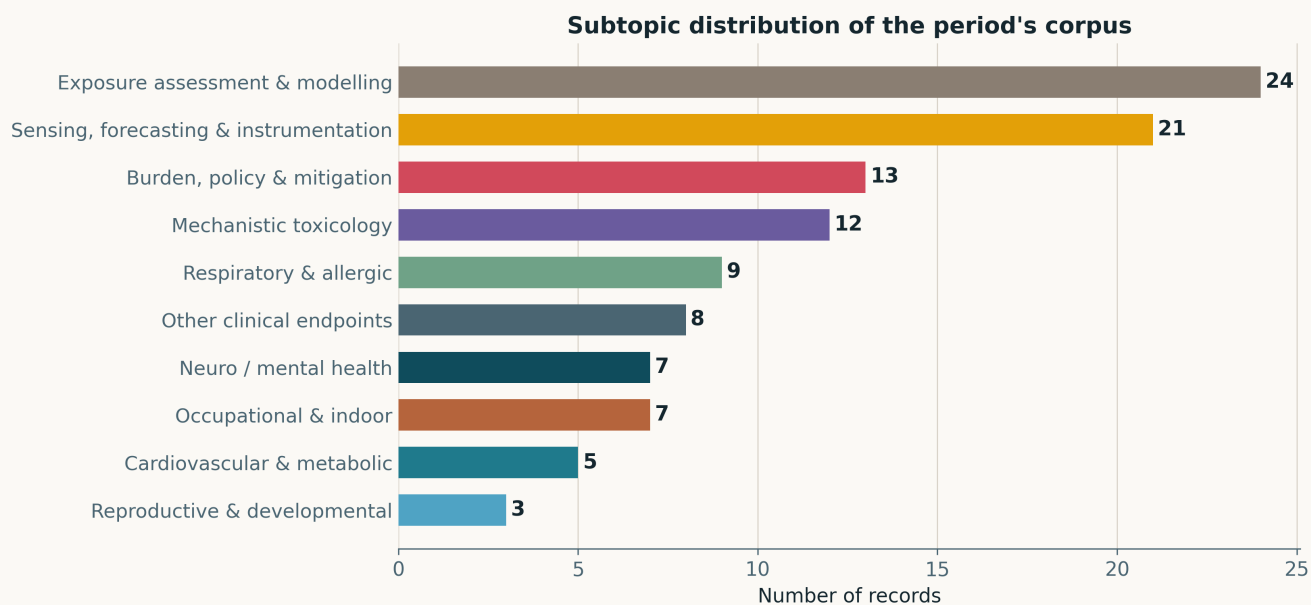


Fig. Pooled subtopic distribution, 23–29 August. *Exposure assessment & modelling* (24) and *Sensing* (21) are 41% of the week between them — the lowest measurement-side share in four weeks. *Burden, policy & mitigation* (13) is the largest health-adjacent cluster, ahead of *Mechanistic toxicology* (12) and *Respiratory & allergic* (9). Every one of the ten canonical clusters is occupied, now for five consecutive weeks.

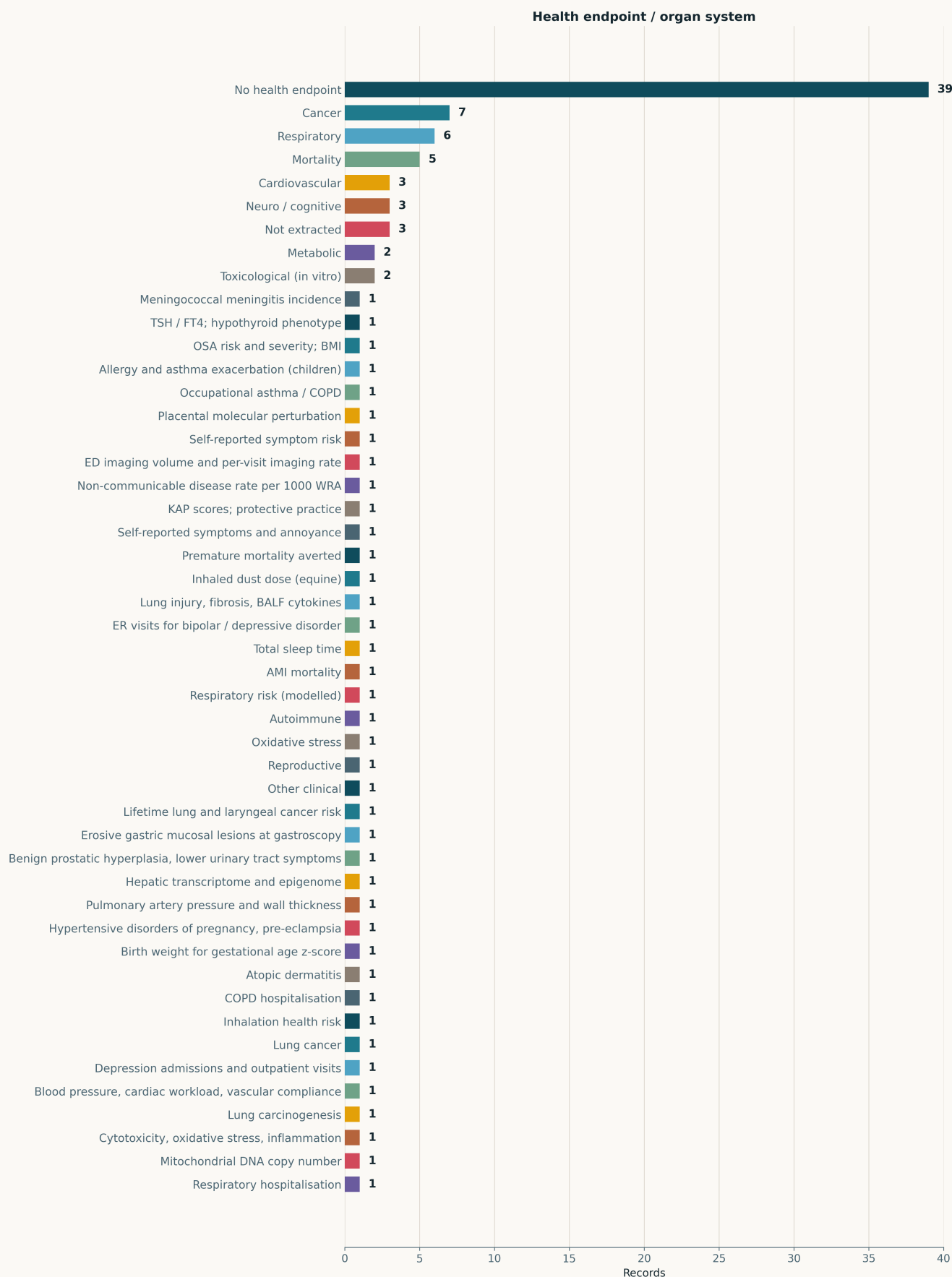


Fig. Health endpoint by organ system across the pooled week. 39 of 109 records carry no human health endpoint — roughly 8 because no abstract was deposited and the endpoint could not be extracted, the rest because they are monitoring, exposure, instrument or aerosol-chemistry work by design. That is 36% against last week's 55%, and the shift is the direct consequence of the abstract-free collapse plus the health-side rebound in the cluster counts. **Endpoint labels remain uncanonicalised**, so the long tail of singleton endpoint strings in this panel overstates endpoint diversity — see the disclosed defect on the venues page.

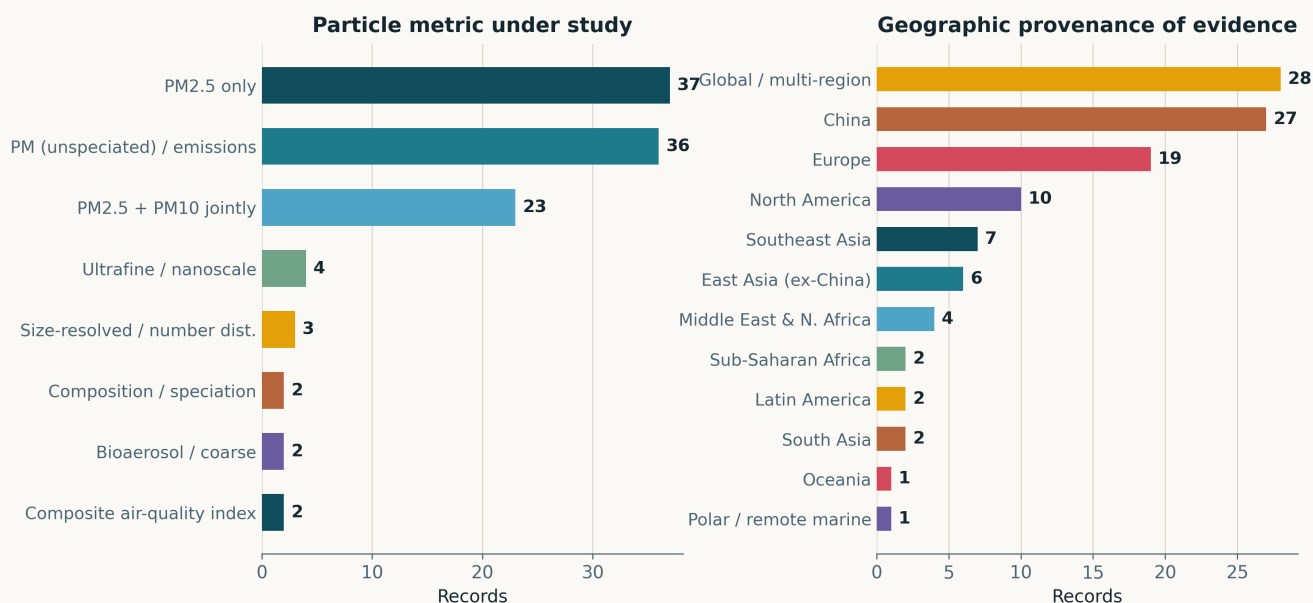
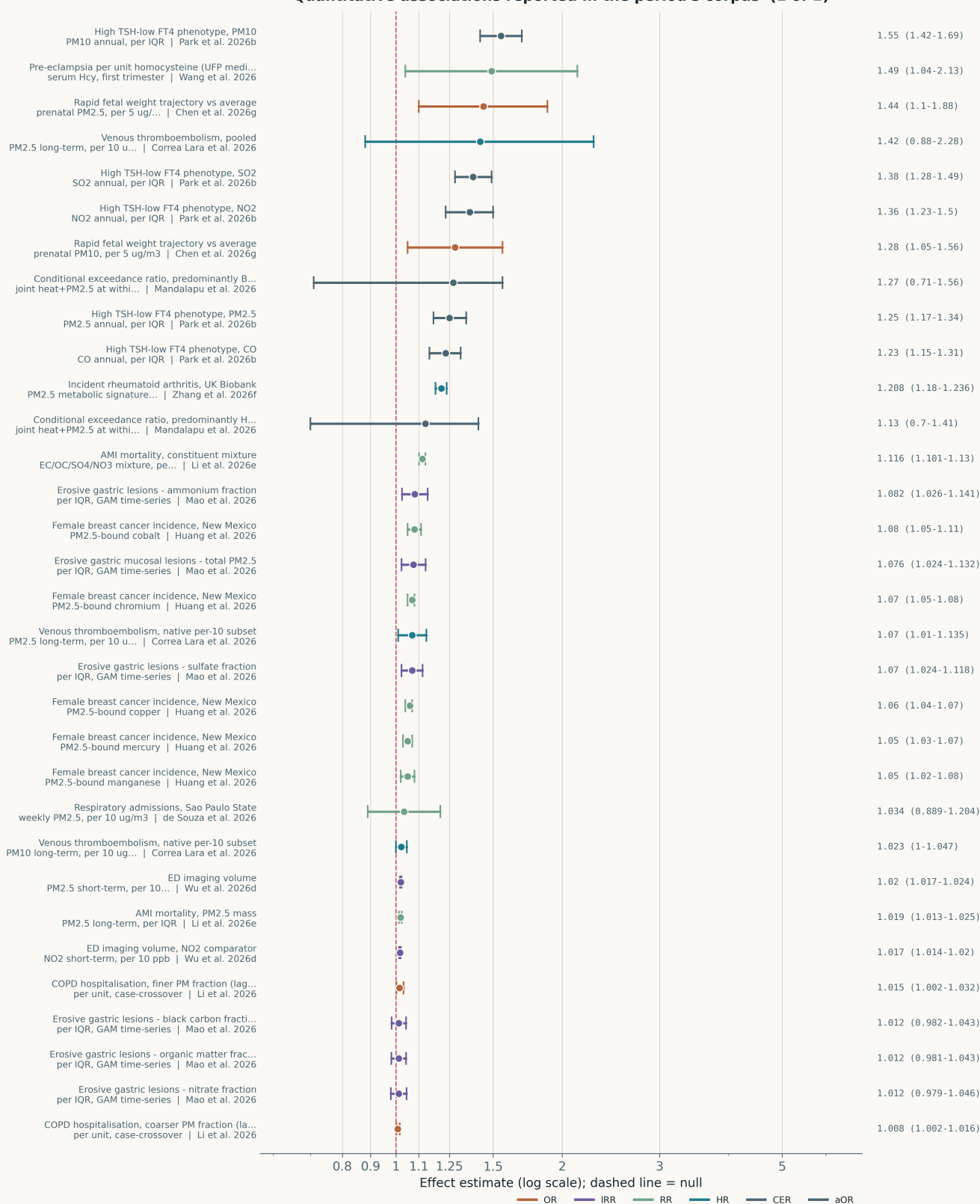


Fig. Left — particle metric across the pooled week. PM_{2.5} alone is the exposure in 37 records, unspeciated PM or emissions in 36, and PM_{2.5} with PM₁₀ jointly in 23. **Ultrafine or nanoscale particles are the explicit subject of 4 records and size-resolved number distributions of 3 more** — 7 of 109, double last week's 2 of 112 and the highest this watch has recorded, though still an order of magnitude below what deposition physics would justify. Only 2 records carry a *composition / speciation* label, which is the standing grouper limitation: the week's constituent-resolved epidemiology bins as *PM_{2.5} only*. Right — geographic provenance: **multi-region syntheses (28) edged past China (27)** for the first time in this watch, with Europe 19 and North America 10. Sub-Saharan Africa contributes 2, both on 29 August, after an eight-issue absence.

Quantitative associations reported in the period's corpus (1 of 2)



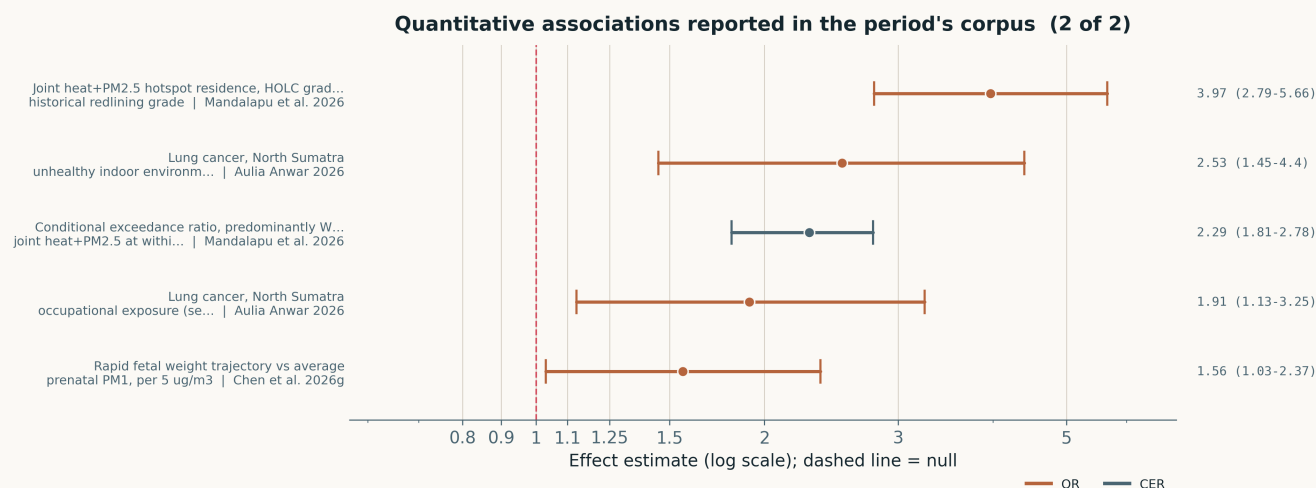


Fig. All 37 pooled effect estimates, split across two panels and plotted on a common logarithmic axis. Exposure contrasts are heterogeneous and the estimates are not mutually comparable — per-IQR, per-10 $\mu\text{g}/\text{m}^3$, per-SD, quartile contrasts and binary residential exposures all appear — so read direction and precision, never relative magnitude, and no meta-analytic summary is computed or implied. Two internally comparable sub-sets exist and are worth separating from the rest. The five Park et al. thyroid estimates (29 Aug) share one outcome, one cohort and one per-IQR contrast, so their ordering is internally consistent — but they come from a single collinear CMAQ field entered in separate single-pollutant models, so the ordering is not a potency comparison. The two Li et al. AMI estimates (28 Aug) are *not* on a common contrast: 1.019 is per IQR of total mass, 1.116 per quartile of a four-component mixture, and the visual gap between them is partly a contrast artefact.

Life-course windows addressed this period (records may span several stages)

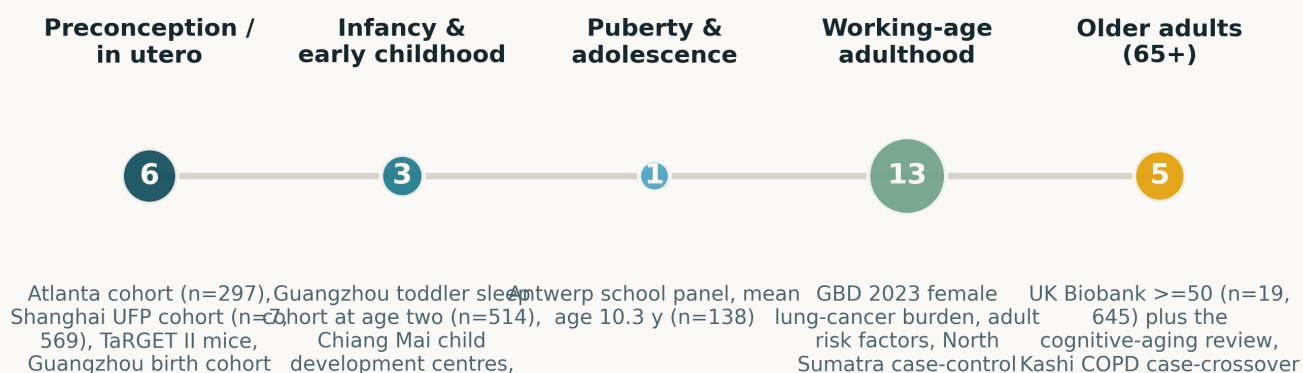


Fig. Pooled life-course coverage. Per-stage counts and annotations are summed and merged across the seven daily issues rather than inherited from any one of them. The working-age and 65+ windows carry this week — CHARLS, the Korean screening cohort, the Stockholm survey — reversing last week's in-utero and early-childhood concentration, consistent with *Reproductive & developmental* falling to 3 records.

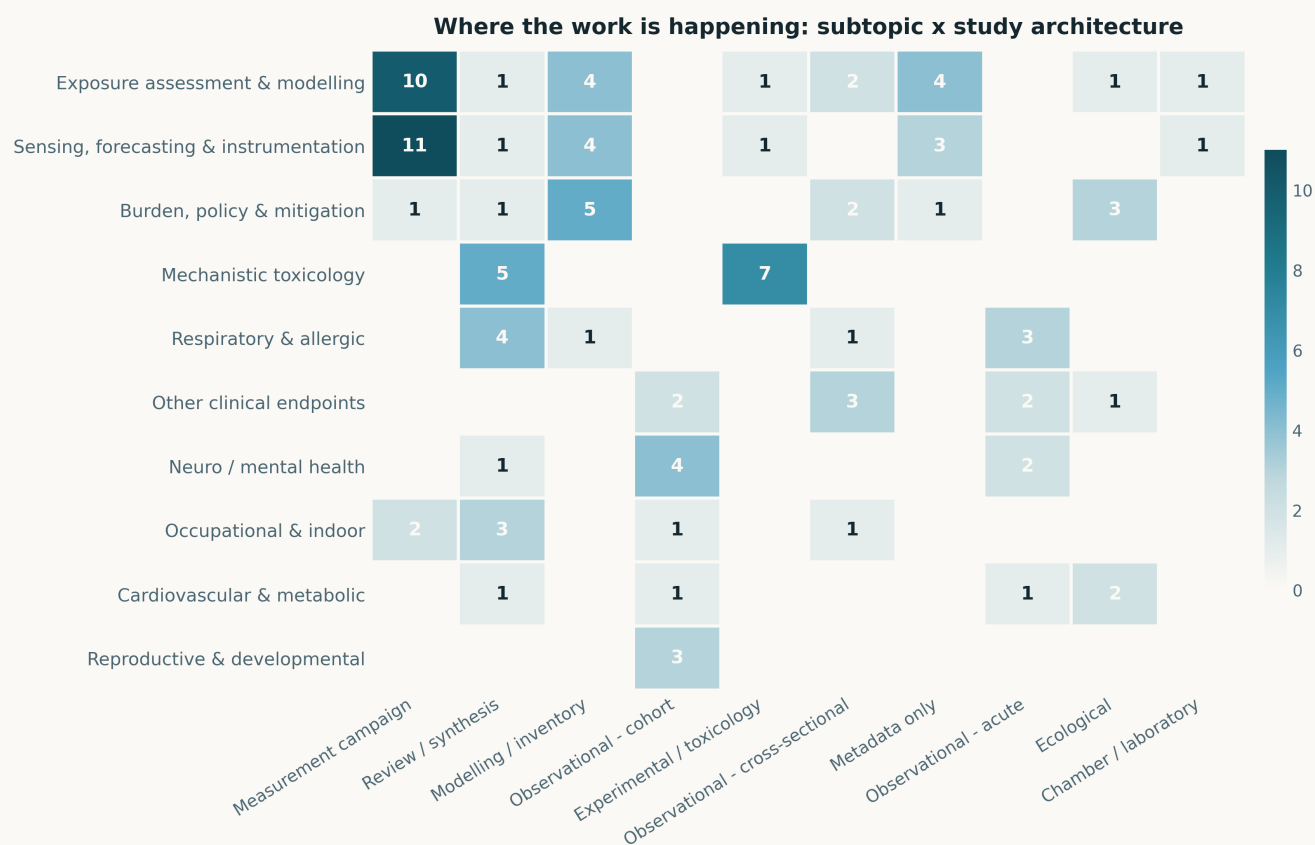


Fig. Subtopic × architecture over the pooled week. **The structural gap this watch has reported every week is visible again:** the *Sensing* and *Exposure* rows contain almost no cells that also carry a health endpoint, and the cohort column contains almost no cells that measured their own exposure. Instrumentation and epidemiology remain methodologically separate populations of paper. The *Metadata only* column, which spanned most subtopic rows last week, is now a thin strip.

Venues, provenance and what the pipeline did

Venues. *Atmos Chem Phys* and MDPI *Toxics* led with 7 records each, then *Environ Pollut* and *Environ Health (Wash)* with 5; *Zhongguo Fei Ai Za Zhi*, *Atmosphere* and *Front Public Health* contributed 4 each. **Six preprints** entered the week (Research Square 3, medRxiv 3) — 5.5% of the corpus. The *Toxics* count is a single volume release on 27 August rather than a sustained venue signal.

Author clusters. By surname the week is dominated by the expected Chinese high-frequency names — Chen (7 records), Zhang (6), Wang (4), Li (4), Yang (3), Wu (3) — but these are *not* research groups: the corpus stores a short label, not an affiliation or an ORCID, so surname frequency here measures nothing beyond the geography already reported (China 27). **No genuine author-cluster analysis is possible with the current record schema**, and this rollout does not attempt one. Adding a first-author affiliation field is the cheapest fix and is logged as an open item.

Two data defects were found by this rollout and are disclosed. First, **two records were published twice** — article numbers 2026.101.13 and 2026.102.16 under the 10.3779/j.issn.1009-3419 prefix, both *Zhongguo Fei Ai Za Zhi* — appearing in the 24 August issue and again on 25 August with differently-cased titles. The DOI leg of `seen.json` should have caught this; it did not, which points at a case-normalisation or write-ordering gap in the same-run screen. The pooled corpus is therefore **109 records over 107 unique DOIs**, and every count in this issue is on the 109 figure. Second, **five short-label collisions** were created within this period — two records each sharing `Chen et al. 2026e`, `Chen et al. 2026f`, `Li et al. 2026d`, `Wu et al. 2026c` and `Zhang et al. 2026f`. Because the short label is the only identifier in the digest and register, a collision makes two different papers indistinguishable. All five were repaired before this rollout was built and the 28 and 29 August issues were recompiled and re-shipped with the corrected labels. **Three older collisions remain in place** (`Wang et al. 2026`, `Yang et al. 2026b`, and `Li & Li 2026`, the last being the duplicate above) and are not retro-fixed, because rewriting a published issue is worse than carrying a documented defect.

Harvester state. OpenAlex returned HTTP 429 or 403 on every run of this period and should be treated as removed from the pipeline rather than degraded. Europe PMC's `CREATION_DATE` leg returned **zero** on both 28 and 29 August, which is new — it was returning archival back-scans in volume last week. arXiv contributed nothing dated in window on any day of the period. Crossref by-ISSN carried the instrumentation axis almost single-handedly: all three *Atmos Chem Phys* single-particle papers of 28 August, which PubMed structurally cannot index, arrived through that leg alone. **Twenty-three design labels and two country mappings** (*Slovakia*, *Italy*) were added to `plots.py` during the period, each in the same edit that wrote them onto a record, per the 22 August guard; one further label was found unmapped by this rollout and fixed here.

Trial watch. `state/trials.json` carries **23 registrations**, last refreshed 22–26 August. Two entered in that window: **NCT05338242** (Monitoring Environmental Exposures and Behavioral Change; interventional, completed, $n = 30$) and **NCT06247059** (Reducing Respiratory Virus Transmission in Bangladeshi Classrooms; **withdrawn**, $n = 0$). The withdrawal is the more informative of the two: classroom filtration trials are exactly the interventional evidence this corpus keeps not containing, and this one did not start. Of the 23 accumulated registrations, 21 are interventional and **air cleaners or HEPA filtration are the intervention in 9 of them** — the registry pipeline is far more interventional than the published literature this watch sees.

Open gaps

- ▶ **Nobody measured the exposure they analysed.** Of the 36 tier-A records this week, every human-health one inherits a modelled exposure surface — CMAQ, satellite fusion, land-use regression or a reanalysis product — and none reports that model's error at the spatial scale of its own analysis. **Fifth consecutive week.** The one partial exception cuts the other way: Wongsasuluk et al. (28 Aug) *did* measure their own exposure, with five low-cost DustBoy units in a road tunnel, and report no co-location against a reference monitor — so the risk estimates inherit an uncorrected nephelometric bias in the highest-humidity, highest-loading regime there is.
- ▶ **Constituent-resolved epidemiology still has no constituent-resolved monitoring behind it.** The week's two strongest health results decompose $PM_{2.5}$ into components taken from chemical transport or satellite products. Zhang et al. (28 Aug) *did* measure composition directly, at single-particle resolution, and has no health endpoint. The two halves of the argument were published on the same day and do not cite each other.
- ▶ **Zero intervention studies in 109 records**, against 21 interventional registrations sitting in the trial watch. The publication lag between the two is now the single most quantifiable gap in this corpus.
- ▶ **Ultrafine coverage doubled and is still negligible.** 7 of 109 records (4 explicitly ultrafine, 3 size-resolved), up from 2 of 112. Best week on record, and still under 7%.
- ▶ **Africa: 2 records in 109, both on the last day of the week.** The preceding eight issues carried none. Both are useful — a West Rand time-activity survey reporting indoor fractions roughly 10 percentage points below USEPA defaults, and a meningitis-belt dust review — and neither involves an African-sited monitoring network.
- ▶ **Endpoint canonicalisation, open since 15 August.** `ENDPOINT_CANON` still does not collapse synonymous endpoint strings, so `f2b`'s tail is inflated and the “no health endpoint” count in this rollout (39) is assembled by rule in the caption rather than read off the figure.

Paper-level digest — the week by subtopic

Every record that appeared in a daily issue keeps that issue's critical summary verbatim; nothing here is re-written or re-assessed. Records surfaced without an abstract are shown as such. Ordered by subtopic, then by the date of the issue that first carried the record.

Sensing, forecasting & instrumentation

21 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (21 OF 21 RECORDS IN THIS SUBTOPIC)

Alharbi et al. 2026 Atmosphere

Morphology, surface chemistry and mineralogy of coastal PM_{2.5}, Jeddah

Three coastal sites over summer 2017 characterised by SEM, EDS mapping, XPS and XRD, linking bulk concentration (**22.2, 18.9 and 14.2 $\mu\text{g}/\text{m}^3$** north to south) to particle morphology, surface oxidation state and crystalline phase. The authors are unusually candid about their own artefact: **borosilicate glass-fibre substrates dominate the SEM field**, so aerosol is resolved only as discrete sub-micrometre particles and agglomerates decorating the fibres, and the morphological interpretation is framed accordingly. That admission is worth more than the result — filter-substrate choice constrains what electron microscopy of a PM sample can establish, and most papers in this genre do not say so.

doi: 10.3390/atmos17090830

Anon et al. 2026 Atmosphere

Metrological consistency of the EN 12341 gravimetric reference method

National interlaboratory comparison run by ISPRA at Castel Romano with **18 Italian regional environmental agencies** sampling side by side, motivated by the absence of Certified Reference Materials for PM and by known sampler-design discrepancies. Systematic between-laboratory deviations were driven mainly by **relative humidity and precipitation** acting on PM thermodynamic and mass stability, with a minor filter-matrix effect. Because gravimetry is the only traceability source for automated measurement systems, the bias propagates into every AMS calibrated against it — and Directive (EU) 2024/2881 tightens AMS residual uncertainty budgets sharply from 2030, at concentrations that are falling. No numeric bias magnitudes appear in the abstract, which is the main gap; the full text is needed to size the effect. Crossref lists no first author.

doi: 10.3390/atmos17090828

Chen et al. 2026e Atmos Environ

Aerosol and VOC pollution characteristics along the lower Yangtze, from shipborne observation

Metadata-only deposit. Shipborne mobile measurement of aerosol and VOCs along an inland waterway corridor — a platform class this watch sees rarely and one with a specific methodological problem: the measurement frame moves through the plume it is sampling, so concentrations are Lagrangian-contaminated in a way that fixed-site data are not, and self-sampling of the vessel's own exhaust is a standing artefact that requires wind-relative rejection to control. Whether the authors address either is not determinable from the title. Carried at tier C; the record is retained rather than rejected because inland-shipping emissions are a genuinely under-monitored PM source and the corridor is one of the most heavily trafficked in the world.

doi: 10.1016/j.atmosenv.2026.122331

Eger et al. 2026 Aerosol Air Qual Res

Compact low- and mid-cost sensors for inland-ship exhaust emissions

One year of onshore co-location on the Rhine against reference instruments, screening sensors on noise floor, correlation and response time — criteria chosen because ship plumes are transients, not the slowly-varying urban background most validation studies target. Qualifying devices: **MA350 black carbon** ($R^2 = 0.95$, **slope 0.77**), **Partector 2 Pro ultrafine particle number** ($R^2 = 0.85$), **K30 FR CO₂** ($R^2 = 0.98$), **Alphasense NO** ($R^2 = 0.99$) and **NO₂** ($R^2 = 0.81$). The limitation the authors state plainly is the useful part: sensor noise and the temperature dependence of the electrochemical cells — which they describe as difficult to correct — cut the number of analysable ship peaks. No PM mass sensor cleared the bar, which is itself a result.

doi: 10.1007/s44408-026-00163-w

Fahim & Karim 2026 Res Sq (preprint)

Cost-aware benchmark of time-series foundation models for PM_{2.5} forecasting

Twenty-nine cities, two domains (hourly PM_{2.5}, 2 m temperature), pre-specified protocols with measured energy and a cost-adjusted decision rule, comparing zero-shot foundation models, transfer learning and tuned specialists. **On PM_{2.5} the zero-shot model and the specialist are statistically indistinguishable (MASE 0.662 vs 0.692)**; transfer learning never significantly beats zero-shot at any budget; frequent retraining costs the specialist roughly ten times the measured energy. The methodological contribution is the demonstration that the specialist's temperature advantage is a *perfect-foresight artefact* of feeding it covariates unknowable at forecast time. Unreviewed, and 29 cities is a small sample for a claim about generalisation — but the artefact critique applies to published benchmarks too.

doi: 10.21203/rs.3.rs-10515545/v1

Giannetti et al. 2026 Environ Pollut

Ants as bioindicators of urban traffic pollution

Thirty traffic-graded sites plus four controls in Parma; magnetic susceptibility and SEM-EDS on *Crematogaster scutellaris* exoskeleton and on co-located soil at 5 cm. **Ant magnetic susceptibility tracked traffic intensity at $\rho = 0.825$ ($p < 0.001$)**; **soil did not** ($\rho = 0.167$, $p = 0.346$). On a log scale the relation is linear ($R = 0.66$, $p < 0.001$) with **1.41-fold per additional 1,000 vehicles/day**, ~700-fold across the gradient. SEM-EDS found Fe-oxide particles doped with Zn, Cu and Ti concentrated on the thorax, morphologically consistent with traffic-derived ultrafine PM. The weaknesses are that no slope CI is given, magnetic susceptibility is a proxy for the ferrous fraction only (it says nothing about organic or secondary mass), and nest-density confounding with traffic corridors is not addressed. For distributed sensing the interest is orthogonal: a passive, unpowered integrator with month-scale averaging at effectively zero unit cost, usable as a siting prior for where to put real instruments rather than as a replacement for them.

doi: 10.1016/j.envpol.2026.129010

Kang et al. 2026b J Hazard Mater*Emission characteristics and health impacts of vehicle-originated condensable PM*

Condensable PM from **eight vehicle types** characterised with a purpose-built real-time system (PNU-CPM) plus offline physicochemical analysis, with emission factors derived from the real-time record. Untreated **diesel levels 4 and 5 emitted the most CPM and total PM**; CPM mass fractions were **63.2 % (diesel L4), 8.2 % (GDI) and 58.8 % (LPG 2)**. Organic CPM was ~53.5 % alkanes and ~5.1 % esters; dominant water-soluble ions were SO_4^{2-} and Na^+ . MPPD dosimetry gives higher deposited mass per body weight in younger age groups. Directly relevant to optical sensing: CPM is ultrafine, semi-volatile and forms after the tailpipe, so it is missed by hot-filter source testing and will not follow the scattering or hygroscopic-growth assumptions that low-cost sensor corrections inherit from combustion soot. No on-road fleet measurement — these are dynamometer emission factors.

doi: 10.1016/j.jhazmat.2026.143367

Lefevre et al. 2026 Data Brief*Urban microclimate dataset from low-cost sensor networks, Reunion Island*

A data descriptor for **25 self-built LoRa weather stations** deployed on residential buildings across two contrasting neighbourhoods (Saint-Denis, Le Port), October 2024 to June 2025, at **10-minute cadence** for air temperature, relative humidity, wind speed, global horizontal solar radiation and grey-globe temperature. Station metadata, QC procedures and maintenance logs are published alongside, and incomplete stations are identifiable. **There is no PM channel** — this is a microclimate network, not an air-quality one — so its value here is as a deployment reference: dense neighbourhood-scale LoRa siting, self-built hardware, published QC, and an explicitly stated reuse path for model calibration (ENVI-met, STEVE). Tropical hot-humid deployments are exactly the regime in which PM sensor RH correction fails worst, and this is a rare open record of what such a network's data actually look like.

doi: 10.1016/j.dib.2026.113166

Li et al. 2026c Plants (Basel)*Early warning of cucumber angular leaf spot from airborne pathogen aerosols estimated with PM sensors*

Two-stage greenhouse surveillance in which **low-cost PM sensors act as continuous physical proxies** for suspended pathogen aerosol, and qPCR air sampling is triggered only when an estimated aerosol risk crosses a threshold. Evaluated on **732 plot-days** across spatially independent trials: an Extra Trees model predicted pathogen-aerosol accumulation at $R^2 = 0.884$, and an XGBoost model integrating aerosol load with microclimate forecast the daily disease-index change rate at $R^2 = 0.874$. SHAP identified aerosol concentration and vapour pressure deficit as the primary drivers. The limitation is stated by the authors and is the right one: PM sensors have no biological specificity, and the design compensates by scheduling a specific assay rather than by claiming the sensor sees pathogens. Cross-greenhouse rather than cross-site validation, and no reference-instrument comparison for the PM channel itself.

doi: 10.3390/plants15162510

Marcon et al. 2026 Curr Environ Health Rep*Citizen science in air pollution monitoring: methodological approaches, effectiveness and integration with health research*

Critical review of participatory air-quality monitoring — low-cost sensors, wearables, community co-creation — and its uptake into environmental health research. The review identifies AI-assisted calibration and distributed sensor networks as the methodological frontier and argues that hyperlocal citizen data materially improves spatial resolution and risk communication. Its useful contribution is the ordering of the barriers: **representativeness and policy uptake, not sensor accuracy**, are named as the binding constraints. No quantitative synthesis, no pooled performance figures and no inclusion criteria are reported in the abstract, so it is a framing document rather than an evidence base. For a distributed low-cost network this is the argument that siting design deserves as much engineering attention as the calibration pipeline.

doi: 10.1007/s40572-026-00559-1

Moteki et al. 2026 J Aerosol Sci*Refractive index of fractal-like aggregates from complex forward-scattering amplitudes*

Metadata-only deposit — title and DOI without an abstract. The method named in the title is the single-particle complex forward-scattering amplitude $S(0^\circ)$ approach, previously used to constrain the black-carbon refractive index at $\lambda = 0.633 \mu\text{m}$; the new element is **mass classification of the aggregates** before the optical measurement, which removes the size-distribution term that dominates the uncertainty in bulk retrievals. If it delivers, this is the most directly useful sensing record in the window: every optical PM instrument, from a nephelometer to a \$20 laser scatter sensor, converts scattered intensity to mass through an assumed refractive index, and for soot-dominated aerosol that assumption is the largest single error term. Carried at tier C on the title alone and placed at the top of the abstract retry list.

doi: 10.1016/j.jaerosci.2026.106884

Neuberger et al. 2026 Atmos Chem Phys*Importance of hydrated aerosol particles for aerosol–fog relationships in the Italian Po Valley*

FAIRARI 2021/2022 aerosol–fog observations at San Pietro Capofiume, analysed with κ -Köhler theory and the MIMICA large-eddy simulation. Median hygroscopicity separated by population: **fog residuals $\kappa = 0.45$, interstitial particles 0.40, out-of-fog aerosol 0.34**, the gradient reflecting inorganic enrichment in activated droplets. Excluding hydrated but non-activated particles from the analysis **inflated effective diameter by 81 % (11.6 $\rightarrow$ 21.0 μm) and depressed droplet number concentration by 87 % (97.4 $\rightarrow$ 12.4 cm^{-3})**; hydrated particles supplied **21 % of liquid water content** and drove **36 % of sub-kilometre visibility events without any activation**. The LES leg shows fog prediction hinging on the largest dry particles. The limitation that binds is site specificity — one heavily polluted valley over two winters, with a κ distribution shaped by Po Valley nitrate that will not transfer to a marine or biomass-burning regime. For distributed low-cost sensing the transferable point is that the hydrated/activated distinction is a *composition* problem, so an RH correction fitted at one site during one season inherits that site's κ and no more.

doi: 10.5194/acp-26-11991-2026

Roska et al. 2026 Environ Sci Technol*Beyond targeted calibrations: comprehensive quantification of oxygenated organic compounds in urban atmospheres*

A collision-induced-dissociation voltage-scanning method to derive sensitivities for an ammonium-adduct CIMS, so OVOCs without an authentic standard can still be quantified. Validated in the laboratory, then applied to the **2023 AEROMMA airborne campaign and the 2024 SAPHIR-CHANEL chamber campaign**: coverage rose to **98% of total signal in chamber and 78% in the field**, and adding the newly quantified species raised estimated **OH reactivity by ~5% and SOA formation potential by ~62%**. Coverage degrades with molecular size and oxidation state, and chamber-derived sensitivities are being transferred to ambient air on the strength of trend agreement rather than field standards — the paper says so. The 62% figure is the one that matters downstream: SOA yields used in chemical-transport models that feed exposure surfaces are calibrated against measurements that were missing most of this signal.

doi: 10.1021/acs.est.6c04149

Vivanco et al. 2026 Environ Pollut*Responsiveness of an air quality model to real-world emission changes*

Metadata-only deposit — title and DOI without an abstract, so the design, model, domain and emission perturbations are all unknown at the time of writing. The question is a live one (whether CTMs used for planning actually reproduce the concentration response to observed emission changes rather than only to scenario deltas) and the record is carried at tier C so it is not lost. Flagged for abstract retry.

doi: 10.1016/j.envpol.2026.129031

Wang et al. 2026d Atmos Chem Phys*Perchloric acid drives new particle formation enhanced by dimethylamine, ammonia and sulfuric acid*

Quantum-chemical calculations coupled to the Atmospheric Clusters Dynamic Code compare the nucleation enhancement of DMA, NH₃ and H₂SO₄ for perchloric-acid-based new particle formation, motivated by recent Arctic observations of atmospheric HClO₄. Both DMA and NH₃ bind PA strongly through hydrogen bonding and proton transfer, and **PA nucleates more readily with DMA than sulfuric acid does**; the PA–DMA cluster formation rate exceeds PA–NH₃ even at overwhelming NH₃ excess, with 1:1 PA:DMA clusters controlling growth. Purely computational — no chamber or field verification, and the claimed NH₃ excess of “10–100 orders of magnitude” is a transcription problem in the abstract rather than a physical statement. Relevant as a candidate nucleation channel in clean marine and polar boundary layers where sulfuric acid is scarce, which is where number-concentration instruments see unexplained events.

doi: 10.5194/acp-26-12171-2026

Werderman et al. 2026 Atmos Pollut Res*Laboratory evaluation of Sensirion SEN66, Piera IPS-7100, PurpleAir PIXEL and OPC-N3 low-cost aerosol sensors across aerosol types and concentration ranges*

Metadata only. Crossref by-ISSN sweep; title, authorship (Werderman, Richards, Berkuta, Sousan) and DOI verified, no abstract deposited and no numbers extractable. It is nevertheless the single most on-topic record of the window: a chamber characterisation of *four* current-generation units across aerosol type and concentration range, three of which — SEN66, IPS-7100, PurpleAir PIXEL — have no prior evaluation in this corpus, against the OPC-N3 as the established reference point. The same last author’s 2021 four-sensor evaluation established that environmental and occupational settings need separate calibration factors, so the plausible contribution here is the aerosol-type dependence of that factor for the newer optics. **Highest-priority full-text retrieval on the next run**; nothing is asserted about its findings until then.

doi: 10.1016/j.apr.2026.103185

Wongsasuluk et al. 2026 Environ Sci Pollut Res Int*PM_{2.5} exposure and respiratory health risk for vulnerable populations in an urban underpass: a predictive MLR approach, Chiang Mai*

Five DustBoy low-cost stations along the Don Kaew road tunnel, August 2022 to July 2023, yielding **385,622 records** across PM_{2.5}, PM₁₀, CO, CO₂, NO₂, SO₂, O₃, TVOC, CH₂O and meteorology. A stepwise multiple linear regression stratified by weekday/weekend and day/night gives a high reported goodness of fit, and a comparative risk assessment converts predicted concentrations into inhaled dose and relative risk against Thai and WHO exposure factors. The inference-threatening gap is that **no co-location against a reference monitor is reported**: a semi-enclosed tunnel is a high-humidity, high-loading environment, precisely where nephelometric low-cost sensors over-read most, and every downstream risk number inherits that bias uncorrected.

doi: 10.1007/s11356-026-38171-3

Xiong et al. 2026 Atmos Chem Phys*In situ real-time determination of SO₂ photochemical oxidation in nanoscale sea salt aerosols based on dark-field microscopy*

Planar-waveguide dark-field microscopy tracks hygroscopic growth factors of single NaCl nanoparticles under UV irradiation and SO₂, with the Zdanovskii–Stokes–Robinson rule converting growth to composition and hence to reaction rate. Reported first-order rate constant **0.6523 h^{−1} for 100 nm NaCl**, with a **non-monotonic size dependence** — rising from 50 to 200 nm, falling above 200 nm — attributed to competition between surface curvature and specific surface area, and cross-checked against transition-state theory and a double-film mass-transfer treatment. Single salt system under controlled UV; the extrapolation to ambient mixed-composition marine aerosol is not demonstrated. Methodologically this is the interesting part: an optical single-particle technique that yields kinetics without collection, at sizes below what most PM instrumentation resolves at all.

doi: 10.5194/acp-26-12183-2026

Yu et al. 2026b Indoor Air*Machine learning-based prediction of occupant symptom risk using indoor and outdoor environmental quality data in university dormitories*

Seven monitoring periods across two Milan student residences, May 2024 to June 2025, pairing continuous indoor and outdoor environmental quality data with occupant symptom surveys. From **74 occupant samples** and 131 candidate predictors (reduced per symptom to 3–40 by permutation importance), seven binary symptom classifiers were fitted under leave-one-period-out cross-validation. Performance is weak and honestly reported: **mean AUC 0.595, range 0.398–0.705**, four of seven above 0.6; the higher combined accuracy figures appear only after dropping the two least specific symptoms. With $n = 74$ against 131 candidate predictors the feature-selection step is almost certainly overfitting even under LOPO-CV. The transferable finding is the feature ranking: **outdoor statistics and distributional shape features beat mean indoor concentrations**, which argues directly against summarising a sensor time series by its mean.

doi: 10.1155/ina/8230467

Zhang et al. 2026c Atmos Chem Phys*Buffering of atmospheric nanoparticle growth by temperature-dependent shifts in molecular composition, volatility and diffusivity*

A kinetic multiphase treatment applied to previously unreconciled field and chamber nanoparticle growth-rate data. The standing puzzle is that observed growth rates are smaller than theory predicts and *less* sensitive to condensable vapour concentration than theory predicts. The resolution offered is two antagonistic temperature dependencies: **slow surface-to-bulk transport limits vapour uptake by semi-solid, low-diffusivity particles, while Clausius–Clapeyron shifts in the volatility distribution enhance growth at low temperature and low concentration**. Netted out, they buffer the vapour-concentration dependence, which is why measured growth rates converge on a few nm/h across widely varying conditions. No new measurement is reported — this is a reinterpretation of existing datasets, and its persuasiveness rests on the volatility-distribution parameterisation rather than on independent validation. The relevance here is upstream of sensing: the 50–100 nm photochemical mode that Zhong et al. flagged yesterday as invisible to mass-based assessment is produced by exactly this growth process, and a mechanism that makes its rate insensitive to precursor loading also makes it insensitive to the emission controls that reduce PM_{2.5} mass.

doi: 10.5194/acp-26-12037-2026

Zhang et al. 2026h Atmos Chem Phys*Contrasting nighttime heterogeneous and daytime photochemical aging drive the optical evolution of black carbon*

Integrated SP2 and single-particle aerosol mass spectrometry over an urban Shenzhen campaign, mean refractory BC $1.2 \mu\text{g m}^{-3}$, core mass median diameter 155–170 nm. The coating-to-core mass ratio rises continuously through the diurnal cycle by two mechanistically distinct routes — nighttime heterogeneous coating growth with enhanced nitrate signal, and daytime photochemical aging with rapid nitrate accumulation followed by sulfate and oxidised organics. Both converge on similar mass absorption cross section enhancement at 532 nm: $\sim 0.8 \text{ m}^2 \text{ g}^{-1}$ **overnight**, $\sim 1.0 \text{ m}^2 \text{ g}^{-1}$ **by day**. No uncertainty is given on the MAC enhancements and it is a single city over one campaign, so the magnitudes should not be transported. Directly relevant to distributed sensing: any optical channel with a fixed absorption or scattering assumption re-expresses this aging cycle as concentration drift with a diurnal period.

doi: 10.5194/acp-26-12197-2026

Exposure assessment & modelling**24 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (24 OF 24 RECORDS IN THIS SUBTOPIC)

Agpaoa et al. 2026 Environ Monit Assess*Occurrence, temporal variation and meteorological drivers of airborne fungal spores in Laoag City, northern Philippines*

First aeromycological record for a monsoon-influenced coastal site in northern Luzon: 216 sampling days, March to November 2022, cumulative index **190,096 spores m^{-3}** , dominated by *Cladosporium* and Fusaroid-like types. Wet-season sampling carries most of the cumulative load but the highest daily concentrations fall in the dry season, and diurnal peaks sit at 00:00–06:00. Positive associations with relative humidity and rainfall, negative with temperature, and no significant wind association. Single site, single year, morphology-based identification only, and no PM co-measurement — so the bioaerosol fraction cannot be placed within a mass-based PM budget. Included because overnight spore peaks are a documented confounder for optical particle counters in humid climates.

doi: 10.1007/s10661-026-15814-z

Anon et al. 2026b Environ Int*Condensable particulate matter and severe winter $\text{PM}_{2.5}$ in remote Canadian mountain communities*

Metadata-only deposit — title and DOI without an abstract. The topic is directly relevant and under-covered: condensable PM forms downstream of the stack as vapour cools, is invisible to filter-based source testing conducted at temperature, and in subzero valley-inversion conditions can dominate ambient $\text{PM}_{2.5}$ in communities with no industrial source of any size. If the finding is what the title implies, it also bears on optical sensing — semi-volatile condensable material has different scattering and hygroscopic behaviour from the combustion soot most correction factors assume. Carried at tier C and flagged as the top abstract retry.

doi: 10.1016/j.envint.2026.110472

Anon et al. 2026c Atmos Pollut Res*Source-specific $\text{PM}_{2.5}$ oxidative potential in a national shipbuilding complex*

Metadata-only deposit. The title asserts that metal processing and combustion dominate source-apportioned oxidative potential, which if supported is another instance of the recurring result that OP-weighted and mass-weighted source contributions rank differently. No apportionment method, OP assay, site count or period is available. Tier C, second on the abstract retry list.

doi: 10.1016/j.apr.2026.103189

Chen et al. 2026f Toxics*Source-specific oxidative potential of $\text{PM}_{2.5}$: water-soluble metals via DTT and interpretable ML*

Winter and summer ambient $\text{PM}_{2.5}$ in Xi'an assayed for DTT oxidative potential, with XGBoost–SHAP attribution over composition and receptor-model source contributions. **Volume-normalised OP (DTT_v) was markedly higher in winter**, and although water-soluble metals are a small fraction of $\text{PM}_{2.5}$ mass, **potassium and manganese dominated the OP attribution** — a biomass-burning tracer and a redox-active transition metal. Source apportionment separates the two seasons cleanly: dust drives winter OP through mass loading, traffic drives summer OP through intrinsic per-mass toxicity. The abstract reports no numeric DTT_v values, confidence intervals or model performance, which is a real gap for a paper whose central claim is quantitative attribution; SHAP importance is also not a causal decomposition and correlated constituents will trade rank. The qualitative conclusion — OP tracks composition and source, not mass — is the second independent instance of it this week.

doi: 10.3390/toxics14080646

Cui et al. 2026 Environ Pollut*Daily 1 km satellite-based $\text{PM}_{2.5}$ exposure estimates in Bangladesh, 2020–2025*

Two-stage machine learning over one of the most monitoring-sparse high-burden regions in the world. Stage one reconstructs MAIAC aerosol optical depth, of which **more than 60% of potential retrievals are missing**, with LightGBM; stage two feeds the gap-filled AOD plus meteorological, composition and geospatial predictors into an attention-based TabNet to estimate ground $\text{PM}_{2.5}$. Independent test performance $R^2 = 0.95$, **RMSE $17.41 \mu\text{g/m}^3$** , **MAE 9.38** , against a national annual mean of **$68\text{--}75 \mu\text{g/m}^3$** and winter peaks above 130. The limitation is structural: with almost no ground reference network, the held-out set is small and drawn from the same sparse monitors used in training, so $R^2 = 0.95$ describes interpolation skill more than transferability, and the DHS linkage is division-level and ecological.

doi: 10.1016/j.envpol.2026.129051

Hopke et al. 2026 Environ Monit Assess*A $\text{PM}_{2.5}$ magnitude-rank index for summarizing annual concentration-frequency structure*

Introduces the $\text{PM}_{2.5}$ Magnitude–Rank Index — the largest integer k such that at least k days in a year have daily mean $\text{PM}_{2.5} \geq k \mu\text{g/m}^3$ — as a threshold-free joint summary of magnitude and frequency. Applied to **918 qualifying MSA-years across 96 US metropolitan areas, 2013–2022**, PMRI ranges 6–38 (median 16) and correlates with the annual mean at Spearman $\rho = 0.83$, the annual 98th percentile at 0.95, and days $\geq 15 \mu\text{g/m}^3$ at 0.96. That last figure is the honest problem with the metric: at $\rho = 0.96$ against an existing exceedance count it adds very little information, so its case rests on being policy-threshold-free and intuitively readable rather than on discriminating power. No health-outcome validation is attempted.

doi: 10.1007/s10661-026-15858-1

Isworo & Oetari 2026 Environ Sci Pollut Res Int

Risk-integrated industrial air quality: Gaussian dispersion linked to a health-weighted pressure index

Gaussian-plume dispersion for a Central Java textile facility combined with a hazard-quotient-normalised Atmospheric Environmental Pressure Index, using an emission inventory, ERA5 2015–2024 meteorology and six receptor measurements. The headline is a compliance paradox worth stating: **modelled concentrations meet Indonesia's national standards while PM_{2.5} exceeds the WHO 2021 guideline**, and PM_{2.5} and PM₁₀ dominate the index. AEPI values (0.06 baseline, 0.08 predicted) sit in the authors' own "very low" band, which makes the index insensitive at exactly the concentrations where the health argument bites — a scaling problem the paper does not confront. One facility, six receptors, no validation of the plume model against the measurements. Tier C: the framework is transferable, the result is not evidence.

doi: 10.1007/s11356-026-38159-z

Kalumbi et al. 2026 J Expo Sci Environ Epidemiol

Age-specific time-spent patterns among residents living near gold mine tailings storage facilities in the West Rand, South Africa

Cross-sectional structured-questionnaire survey of **330 households in Kagiso**, West Rand, resolving indoor and outdoor time across five age categories with Kruskal–Wallis tests and Bonferroni-adjusted Dunn's comparisons. Indoor time runs **66.6–84.8% of a weekday and 76.0–82.7% of a weekend day** (18.28±3.02 h weekdays, 19.00±1.99 h weekends); young children and older adults are indoors most, and weekend indoor time rises significantly for children, adolescents and adults but not for the two extreme age groups. The headline for exposure modelling is that indoor time is **nearly 10 percentage points below USEPA Exposure Factors Handbook defaults**. Self-reported recall with no diary or wearable validation, and a single district, so the magnitude is soft — the direction of the bias in imported default time budgets is the usable result.

doi: 10.1038/s41370-026-00972-6

Kang et al. 2026 Atmos Environ

Characteristics and dry deposition fluxes of particulate parent, alkyl, and nitro-polycyclic aromatic hydrocarbons in the Bohai Sea Strait, China

Metadata only. Crossref sweep, no abstract deposited. Nitro-PAHs are among the more directly mutagenic particle-bound species and are under-measured relative to the parent compounds, and a marine-strait dry-deposition budget is a receptor-side measurement rather than an emission inventory — but no flux, no concentration and no sampling period are extractable from the deposited record. Flagged for retry, low priority relative to the wind-correction paper.

doi: 10.1016/j.atmosenv.2026.122322

Kim et al. 2026 Atmos Environ

Application of machine learning-based wind corrections to coupled WRF–CMAQ simulations: a comparison of dynamic assimilation and static replacement

Metadata only. Crossref by-ISSN sweep; title, authorship and DOI verified, no abstract deposited. The comparison it names — injecting an ML-corrected wind field into CMAQ dynamically during the run versus statically replacing the WRF field beforehand — is a live methodological question for anyone producing exposure surfaces, because wind error is the dominant transport error in urban CTM output and the two strategies have very different mass-conservation properties. Nothing is asserted about which won. Scholar Gateway cannot reach it (corpus ends May 2026) and Consensus has not indexed it. Second-priority abstract retry behind Werderman et al.

doi: 10.1016/j.atmosenv.2026.122324

Leibensperger et al. 2026 Atmos Chem Phys

Sea surface warming suppresses primary sea spray aerosol number production

Controlled wave-breaking experiments in the Scripps Ocean–Atmosphere Research Simulator, which reproduces wind, waves and sea surface temperature together. Raising SST from 2 to 23 °C **suppressed subsurface bubble concentration by 1.5×, sea spray aerosol number by 3× and SSA number flux by 3×**, with a non-monotonic SST dependence in the supermicron fraction. This is a single simulator with one seawater matrix and no biological aging, so absolute fluxes are not transferable — the value is the sign and rough magnitude of the SST dependence, which the field has not agreed on. Relevance here is indirect but real: marine background number concentration is the baseline against which coastal low-cost network anomalies are read.

doi: 10.5194/acp-26-12211-2026

Li et al. 2026d Environ Pollut

Surface PM transport under a closed earth-air model

One year of continuous in-situ monitoring in the semi-enclosed basement of the Mogao Grottoes conservation department, testing whether **earth-air vertical movement** driven by atmospheric-pressure fluctuation controls surface PM independently of wind and human activity. Structural equation modelling gives **CO₂ as a tracer for EVM-driven PM dynamics**, with path coefficients of **0.452 (PM_{2.5}), 0.470 (PM₁) and 0.386 (PM₁₀₀)**; ²²²Rn regulates PM positively but far more weakly, and relative humidity is strongly inhibitory. PM_{2.5} and PM₁ are strongly affected by EVM, PM₁₀₀ only weakly. Single site, single enclosure, and SEM path coefficients on autocorrelated one-year time series overstate precision badly. The mechanism claim is interesting for indoor and heritage monitoring; the effect sizes should not be transported.

doi: 10.1016/j.envpol.2026.128983

Mainka & Mainka 2026 Preprints.org

Urban cycling versus PM_{2.5} exposure: does the benefit outweigh the risk?

Scenario analysis of **162 real commuting trips by two cyclists** in Gliwice across heating and non-heating seasons, with personal PM_{2.5} measured by low-cost sensor and inhaled dose reconstructed from concentration, trip duration and cycling-specific ventilation. Median trip 33.0 min/day. The median physical-activity relative risk was **0.836** (a 16.4% all-cause mortality reduction) against a median PM_{2.5} relative risk of **1.00182** (0.182% increase), for a **net RR of 0.840 (IQR 0.815–0.879)**; **160 of 162 trips retained net benefit** under sensitivity analysis. Two hard limits: **n = 2 commuters**, so the exposure distribution is a convenience sample of two routes, and the risk functions are transported from published cohorts rather than estimated. Unreviewed preprint. The useful part is the method — trip-resolved dose from a low-cost sensor is the right exposure object for this question, and almost no one computes it.

doi: 10.20944/preprints202608.1778.v1

Mandalapu et al. 2026 medRxiv (preprint)

Joint heat and PM_{2.5} co-exposure across 48 US metropolitan areas

42,304 census tracts, 48 large CBSAs, ~174.6 million residents, summers 2015–2020. The framework combines a composite co-exposure index, a joint exceedance indicator, a conditional exceedance ratio against within-group independence, and an upper-tail dependence parameter estimated both non-parametrically (Capéraà–Fougères–Genest) and with a Gumbel copula, with BCa intervals from a 5,000-replicate block bootstrap. **13.21% of residents of predominantly Black tracts versus 3.33% of predominantly White tracts live in joint hotspots; HOLC grade D vs grade A gives OR 3.97 (2.79–5.66)** after adjustment for contemporary racial composition, poverty, renter occupancy and pre-1960 housing. Magnitudes attenuate under near-surface air temperature rather than surface temperature. Unreviewed; the tail-dependence machinery is the transferable contribution.

doi: 10.64898/2026.08.20.26360970

Mohammed 2026 Build Environ*Spatiotemporal dynamics of atmospheric vulnerability in hot arid cities: introducing the Urban Microclimate Air Stress Index (UMASI)*

Metadata only. Crossref sweep, no abstract deposited, no study city named in the metadata. A composite index combining microclimate and air-quality stress is exactly the kind of construct that needs its weighting scheme inspected before it is used, so nothing is characterised here. Flagged for retry; low priority relative to Werderman et al.

doi: 10.1016/j.buillenv.2026.115166

Moryani et al. 2026 Toxics*Bioaccessibility and speciation of heavy metals in PM_{2.5} road dust*

48 grid-based samples across four functional zones (residential, school, hospital, office) in Karachi (>27 million) and Shikarpur (~0.3 million), with the PM_{2.5} fraction of road dust extracted in artificial lysosomal fluid (**pH 4.5, 37 °C, 24 h**) and speciation modelled in PHREEQC and Visual MINTEQ. **Cr, As and Pb showed moderate inhalation bioaccessibility**; contamination and ecological-risk indices were computed alongside. The design addresses a real gap — total metal content is a poor predictor of the dose that reaches lung tissue — but ALF extraction is a simulation with no in-vivo anchor, and the abstract reports no bioaccessible fractions, no concentrations and no zone contrasts, so nothing quantitative can be lifted. Road dust is also resuspended, not directly emitted, which the pollution indices do not distinguish.

doi: 10.3390/toxics14080704

Mueangsai et al. 2026 Res Sq (preprint)*PM_{2.5} health risk quotient near stone crushing plants, Chonburi*

Community cross-sectional study, 368 residents by stratified random sampling, with PM_{2.5} from two fixed stations near a crushing plant averaged and pushed through the USEPA health risk assessment to a hazard quotient. The design's weakness is structural and fatal to individual-level inference: **two fixed monitors supply a single exposure value for every participant**, so all between-person exposure contrast is assumed away and the HQ varies only through the questionnaire covariates. Carried at tier C. The setting — crushing plants generate coarse, mineral-dominated PM that optical sensors size poorly — is the reason it is here at all.

doi: 10.21203/rs.3.rs-10160540/v1

Rashid et al. 2026 Toxics*Ultrafine particles and mortality: scoping review of evidence and exposure-assessment limits*

PRISMA-ScR scoping review of PubMed and Scopus: **21 of 704 screened articles**, 2007–2025, split **14 short-term** (time-series or case-crossover against central-site particle number counters) and **7 long-term** (land-use regression, chemical transport or machine-learning surfaces). **10 of 14 short-term and 6 of 7 long-term studies reported positive associations**, strongest for respiratory mortality. The finding that matters for this watch is the second-order one: **the strongest associations came from the studies with higher-resolution or source-specific exposure assessment**, a pattern consistent with reduced Berkson-type misclassification rather than with true effect heterogeneity. A scoping review does not pool, so there is no summary estimate and no formal heterogeneity statistic; the exposure-method gradient is observed, not tested. Still the clearest statement to date of why UFP regulation stalls on measurement.

doi: 10.3390/toxics14080664

Reyna-Bello et al. 2026 Environ Monit Assess*Seasonal variability of PM_{2.5}-bound metals and meteorology in an urban-industrial area of northern Mexico*

MiniVol low-volume sampling across four seasons in Saltillo during 2023, with ten metals (Al, Cd, Cu, Fe, K, Mn, Na, Ni, Pb, Zn) by flame atomic absorption and PCA against temperature, humidity, wind, precipitation and pressure. Mass is strongly cold-season weighted — **152.9±23.6 µg/m³ in autumn and 115.3±37.2 in winter against 40.7±6.8 in spring** — consistent with suppressed dispersion, while several metals invert that pattern and peak in spring and summer. The reported metal concentrations are implausibly high for PM_{2.5}-bound species (Cd at 3.13 µg/m³ is orders of magnitude above typical urban values), which points to a units or digestion-blank problem; treat the metal magnitudes as unverified and the seasonal contrast as the usable result.

doi: 10.1007/s10661-026-15856-3

Walker et al. 2026 medRxiv (preprint)*Directly measured noise, PM_{2.5} and night light across Rhode Island*

176 sites, 704 site-sessions over roughly a year, spanning the Providence–Central Falls–Pawtucket urban core and four rural municipalities, with A- and C-weighted noise, PM_{2.5}, night-time illuminance and relative humidity carried through an identical model as a negative control. A-weighted noise rose with disadvantage on six of eight area-level markers (**1.3–1.8 dBA per SD; 6.6 dBA between least and most racially diverse neighbourhoods**) and C-weighted levels followed the same gradient, exceeding their A-weighted counterparts at block-group scale for renter occupancy and vehicle absence. **Short-term PM_{2.5} was roughly an order of magnitude weaker**; humidity showed nothing. The negative-control design is what makes the PM null interpretable rather than merely underpowered. Short-term measurement cannot speak to annual-average disparity, and the authors do not claim it does.

doi: 10.64898/2026.08.20.26360933

Wu et al. 2026b npj Clim Atmos Sci*Global fire activity exacerbates extreme PM_{2.5} pollution*

Metadata-only deposit. The title claim — that fire activity acts disproportionately on the *extreme tail* of the PM_{2.5} distribution rather than on the mean — is the one that matters, because attainment statistics, exposure-response functions fitted on annual means and most burden calculations are all mean-anchored and will understate a tail-loading source. It is also the regime in which low-cost networks are most useful and least characterised: sensor response at several hundred µg m⁻³ of smoke is outside the concentration range of essentially every published co-location study. Carried at tier C pending the abstract; a Nature-portfolio deposit with no abstract in Crossref is unusual and may resolve on retry.

doi: 10.1038/s41612-026-01527-0

Zhang et al. 2026b Environ Sci Technol*Photoinduced intraparticle chemistry drives the rapid atmospheric oxidation of biomass burning organic aerosol*

Chamber isolation of biomass-burning aerosol aging pathways — UV-A irradiation, OH-driven SOA formation, heterogeneous OH oxidation, ozonolysis and evaporation — benchmarked against aircraft measurements of wildfire and prescribed-burn plumes. **UV-A irradiation drove intraparticle aging an order of magnitude faster than gas-phase oxidation**, at an estimated quantum yield of ~0.4% (four in a thousand absorbed photons generate particle-phase radicals). The authors estimate that **over half of oxygenated OA in biomass-burning-influenced regions may originate from photo-oxidised primary OA within the first day**. This inverts the standing assumption that gas-phase oxidation dominates BBOA aging, and it matters for exposure work because oxidation state governs both the oxidative potential of the inhaled particle and the optical properties that low-cost scattering sensors respond to — a smoke plume's sensor bias is not constant along its trajectory.

doi: 10.1021/acs.est.6c02532

Zhang et al. 2026d Atmos Chem Phys*Improved representation of isoprene-derived secondary organic aerosol in CAM6-Chem reveals regional contrasts in its long-term changes over China*

An explicit isoprene-SOA formation scheme implemented in CAM6-Chem with high- NO_x gas-phase epoxide precursors and branch-resolved heterogeneous uptake from both low- and high- NO_x pathways. Against ground observations the update **improves the SOA normalised mean bias over China from -76.7% to -51.6%** and reproduces the compositional structure of four explicit ISOA subspecies. Over 2000–2019 the national mean ISOA trend is weak because regional changes cancel: **the sharpest increase is in Southwest China, where rising biogenic isoprene emission is the leading statistical predictor at a 58.3% regression-based contribution**, with the strongest decrease in Shaanxi–Gansu–Ningxia. The honest reading of the headline number is that a -51.6% bias is still a factor-of-two underestimate — the scheme narrows the gap without closing it, and a regression-based attribution is not a causal decomposition. For exposure work the consequence is directional: any health study using a CTM-derived organic-aerosol surface over China is working with a field that is known-low by roughly half, and non-uniformly so by region.

doi: 10.5194/acp-26-11967-2026

Zhong et al. 2026 Environ Int*Photochemical ultrafine particles pose potential inflammatory risk despite low $\text{PM}_{2.5}$ mass*

Three years of hourly Fuzhou observations; lung-deposited surface area compared against what the local $\text{LDSA}/\text{PM}_{2.5}$ ratio predicts, with predicted inflammation from an animal-derived surface-area dose–response. **6.6% of $\text{PM}_{2.5}$ -attainment hours ($<35\mu\text{g}/\text{m}^3$) carried $7.5\times$ the expected surface-area exposure, and 82% of the predicted inflammatory risk in those hours was undetected by mass.** Median $\text{PM}_{2.5}$ during them was $3.0\mu\text{g}/\text{m}^3$, with the dose–response projecting 27% lung neutrophil infiltration. Machine learning attributes the events to photochemical production of 50–100 nm particles on warm, high-radiation summer–autumn afternoons (O_3 and secondary organic carbon dominate importance). **The limitation that threatens the inference is the dose–response itself:** an animal-derived surface-area relation extrapolated to a human population gives a predicted, not an observed, inflammatory response, and no human endpoint is measured here. Even so, the decoupling of LDSA from mass is measured, not modelled, and it is a direct challenge to the assumption that a dense network of scattering-based mass sensors indexes health-relevant exposure.

doi: 10.1016/j.envint.2026.110469

Cardiovascular & metabolic**5 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (5 OF 5 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026 Indoor Air*Residential incense burning, indoor pollutants and cardiovascular haemodynamics*

Seventy-three adults from the YOUNG TAIWANESE cohort across 60 households, with continuous 24 h indoor $\text{PM}_{2.5}$, PM_{10} , CO , CO_2 , TVOC and formaldehyde alongside non-invasive haemodynamic measurement, analysed by mixed-effects models. Incense-burning households showed higher pollutant concentrations and worse haemodynamics — higher blood pressure and cardiac workload, lower vascular compliance. **The finding that matters is interactive: $\text{PM}_{2.5}\times\text{TVOC}$ and $\text{PM}_{2.5}\times\text{CO}$ carried the effects on myocardial contractility and vascular elasticity**, and in non-incense homes elevated CO_2 acted as an independent stressor. No effect sizes are given in the abstract. $n = 73$ with household clustering is thin for interaction terms, and residual confounding by household behaviour is not addressable here. Implication for instrumentation: a PM-only indoor node is blind to the combination that did the work.

doi: 10.1155/ina/8778576

Correa Lara et al. 2026 Thromb Haemost*Ambient PM and venous thromboembolism: updated systematic review and meta-analysis*

19 observational studies, >4.5 million participants, searched across PubMed, Embase, Web of Science and Cochrane, harmonised to $10\mu\text{g}/\text{m}^3$ and pooled under restricted maximum likelihood with Hartung–Knapp adjustment. Overall long-term: **$\text{PM}_{2.5}$ HR 1.42 (0.88–2.28)**, $I^2 = 97.0\%$; **PM_{10} HR 1.18 (0.85–1.64)**, $I^2 = 96.2\%$. Restricting to studies natively reporting per $10\mu\text{g}/\text{m}^3$: **$\text{PM}_{2.5}$ 1.070 (1.010–1.135)** and **PM_{10} 1.023 (1.000–1.047)**, both at $I^2 = 0\%$. Short-term exposures were null. The sensitivity analysis is the finding — near-total heterogeneity removed by excluding rescaled contrasts — but it is also a smaller, non-random subset, and a subset selected on a reporting characteristic can be homogeneous for reasons other than correctness. Inconsistent VTE case definitions across the 19 studies remain unaddressed.

doi: 10.1055/a-2945-4974

Li et al. 2026e Environ Sci Technol Lett*Sulfate-dominated $\text{PM}_{2.5}$ constituents are associated with increased acute myocardial infarction mortality: a US nationwide difference-in-differences study*

County-level US AMI mortality against $\text{PM}_{2.5}$ and its EC , OC , SO_4^{2-} and NO_3^- constituents, analysed with a causal mixture method inside a difference-in-differences design. Total mass: **RR 1.019 (95% CI 1.013–1.025) per IQR**. Constituent mixture: **RR 1.116 (1.101–1.130) per one-quartile increase**, with SO_4^{2-} carrying weight 0.762. Regionally, an eastern county sees **1.545 (0.656–2.437) excess AMI deaths per 100,000 per year** relative to the west per quartile of mixture, SO_4^{2-} -driven in the east (0.707) and OC -driven in the west (0.485). Exposure is county-level and the outcome is mortality rather than incidence, so this is ecological in structure — the DiD design controls time-invariant county confounding but not time-varying composition-correlated confounders such as regional industrial decline.

doi: 10.1021/acs.estlett.6c00288

Ramírez-Moreno et al. 2026 Med Sci (Basel)*Regional stroke hospitalisation and mortality against long-term $\text{PM}_{2.5}$*

Nationwide ecological analysis of Spain's 17 autonomous communities, 2013–2021, **306 sex-specific region-year observations**, linking INCLASNS stroke indicators to MITECO ambient pollution. Mean age-standardised hospitalisation rate 20.1 ± 5.4 per 10,000. **$\text{PM}_{2.5}$ had the largest positive correlation with age-standardised mortality of any pollutant examined ($r = 0.264$, $p < 0.001$)**, which the authors correctly describe as weak; associations with hospitalisation and with other pollutants were small and inconsistent in sign. **Regional variation in stroke burden was not explained by the available pollution indicators.** Exposure is at region-year resolution against sex-stratified outcomes, so the sex analysis is uninterpretable by construction and the authors say so. Reported here as a well-labelled null; the design cannot support individual-level inference and does not claim to.

doi: 10.3390/medsci14040475

Yang et al. 2026b Blood Press*Endothelial function at low-level chronic residential air pollution*

453 Flemish adults (47.7 % women, mean age 52.8 y) with endothelial function by finger photoplethysmography after 5 min ischaemia, related to residential PM₁₀, PM_{2.5}, NO₂ and black carbon from high-resolution spatiotemporal interpolation. Six-year means were **15.9, 12.8, 14.3 and 1.04 µg/m³** — at or below current EU limits and near the WHO interim range. **Dmax was inversely associated with PM_{2.5}** irrespective of risk-factor adjustment; other pollutant–outcome pairs were consistently inverse but weaker ($p < 0.10$). Association sizes **weakened as pollution fell over 2010–2015** ($p \leq 0.044$), and the spatial gradient in endothelial dysfunction tracked the PM_{2.5} gradient. Both endothelial measures predicted a composite cardiovascular endpoint (HR 0.27–0.49). Modest n , and no confidence intervals are given for the exposure–endothelium associations themselves, which is why none appear in the forest plot. The attenuation-with-decline result is the notable one and is rarely reported.

doi: 10.1080/08037051.2026.2724736

Neuro / mental health**7 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (7 OF 7 RECORDS IN THIS SUBTOPIC)

Beydoun et al. 2026 Ecotoxical Environ Saf*The interplay of air pollution with plasma neurodegenerative markers and metabolome for dementia, Parkinson's disease and all-cause mortality*

UK Biobank, **19,645 participants aged ≥50** with complete proteomic, metabolomic and environmental data; Cox models plus multistate Weibull models for transitions from healthy to dementia, Parkinson's disease and death. Higher NfL predicted all three transitions; **GFAP was specific to dementia, HR 2.65 (95% CI 2.17–3.25)**. On the exposure side the result is largely negative: **PM × NfL interacted with mortality only at the nominal level**, NO₂/NO_x × GFAP interacted *negatively*, and there was little evidence of PM interaction with either marker for dementia. Metabolomic principal components (lipid, amino-acid, energy) did modify the biomarker–outcome relations and interacted with PM for PD and dementia; a branched-chain-amino-acid component attenuated the GFAP–dementia association. The interaction tests are nominal and unadjusted for multiplicity across a large grid, so the positive ones should be treated as hypothesis-generating. Cite this for the biomarker result and for the *absence* of a PM interaction, not for a PM effect.

doi: 10.1016/j.j.ecoenv.2026.120657

Chen et al. 2026i J Occup Environ Med*Gender and season disparities in the influence of short-term PM_{2.5} exposure on emergency room visits for mood disorders in Taiwan*

Time-stratified case-crossover with distributed lag non-linear models over **2,172 emergency visits** for bipolar or depressive disorder in the Chang Gung Research Database, 2010–2020, linked to Ministry of Environment monitoring. Associations appear **exclusively in the hot season and predominantly in women**, with delayed effects at lags 3–5 and cumulative effects over lags 0–1 and 0–3 to 0–5. No effect estimates, confidence intervals or exposure contrasts appear in the abstract, which makes the result unquantifiable as reported. With 2,172 events split by sex and season the subgroup cells are small, and a hot-season-only, women-only finding from a design with no pre-registered subgroup plan is the classic shape of a multiplicity artefact. Carried at tier B on design quality, not on the strength of the reported signal.

doi: 10.1097/JOM.0000000000003848

Hendrickx et al. 2026 Int J Environ Res Public Health*Recent air pollution exposure, children's neurocognition and endothelial function*

138 healthy children (mean age 10.3 y, 55.8 % male) across seven Antwerp primary schools including one open-air school, with indoor and outdoor PM and outdoor NO₂ measured and endothelial and attentional outcomes assessed at three repeated visits. Exposure was resolved at **4 h (direct), 24 h (acute) and 1 week (recent)** windows. **Higher direct indoor PM was associated with reduced microvascular dilatation**, while **selective and sustained attention improved with direct and recent PM exposure** respectively — a sign reversal the authors report without resolving, and which most plausibly reflects the same seasonal or activity pattern driving both classroom occupancy and outdoor concentration rather than a cognitive benefit. No NO₂ associations. Excluding the open-air school shifted results, which on $n = 138$ with seven schools means the school-level clustering is doing real work. Tier B for the vascular finding; the attention result should not be cited without the full text.

doi: 10.3390/ijerph23081070

Ma et al. 2026 Ecotoxical Environ Saf*Day-on-day increases in air pollution and depression morbidity, 239 Chinese cities*

Time-stratified case-crossover on **330,610 hospital admissions and 3,726,390 outpatient visits** for depression, 239 prefecture-level-or-above cities, 2013–2017. The exposure metric is the novelty: air pollution increase between neighbouring days (APIN), the current day's absolute concentration minus the previous day's, rather than level. Per IQR increase in APIN at lag 0–3, admissions rose between 0.59 % (0.06–1.13) for 8 h O₃ and 1.66 % (1.09–2.22) for NO₂, with outpatient visits 0.42–0.85 %; associations were robust to adjustment for absolute concentrations except for ozone. **PM_{2.5}- and PM₁₀-specific estimates are not reported in the abstract**, only the range endpoints, so no PM estimate is carried into the forest plot. The rate-of-change framing is worth borrowing regardless of the endpoint.

doi: 10.1016/j.j.ecoenv.2026.120707

Sakhvidi et al. 2026 Ageing Res Rev*Ambient air pollution exposure and cognitive aging; mediators and possible pathways: a systematic review on the epidemiological studies*

16 studies, 72 unique exposure–mediator–outcome associations, six pollutants (PM_{2.5}, PM₁₀, NO₂, NO_x, black carbon, PM₁ components), with PM_{2.5} in 65% of analyses. Outcomes span memory, processing speed and diagnosed dementia; candidate mediators span depression/anxiety/stress, FEV₁/FVC/PEF, stroke and hypertension, CRP, type 2 diabetes, sleep and neuroanatomical change. **Cardio- and cerebrovascular disease emerged as the most consistent mediator**; mental health, sleep and lung function showed potential but unconfirmed mediation. Reporting quality was judged generally good, but most studies carried some risk of bias in outcome assessment and confounding adjustment. No pooled estimate, and 16 studies across six pollutants and this many candidate mediators is a thin base — the useful output is the map of which pathways have been tested at all, and the observation that almost none have been tested with a formal causal-mediation design.

doi: 10.1016/j.arr.2026.103330

Wang et al. 2026e Sleep Med*Speciated fine particulate matter and self-reported sleep time in middle-aged and older Chinese adults*

19,505 CHARLS participants aged 45+, 2011–2018, with 6-month and 2-year $PM_{2.5}$ and five constituents — BC, organic matter, SO_4^{2-} , NO_3^- , NH_4^+ — from the Tracking Air Pollution in China product, against total sleep time (nocturnal plus napping) via linear mixed models and quantile g-computation. The secondary inorganic ions NO_3^- , SO_4^{2-} and NH_4^+ show the most consistent associations, and the direction is the surprise: highest versus lowest quartile of intermediate-term exposure **increases** total sleep time ($PM_{2.5}$ +0.16 h and comparable for the constituents). Self-reported duration, no actigraphy, and an increase in sleep duration is not obviously a harm — it may index fatigue, reduced outdoor activity on high-pollution days, or reverse behavioural adaptation. Mechanism is not addressed and the finding should be treated as hypothesis-generating.

doi: 10.1016/j.sleep.2026.109229

Wu et al. 2026c Toxics*Early-life PM exposure and toddler sleep: particle size and constituents*

Birth cohort of **514 mother–child pairs** in Guangzhou followed to age two (2017–2018), with PM_1 , $PM_{2.5}$, PM_{10} and $PM_{2.5}$ constituents from high-resolution ground-level grids and sleep by the Chinese Brief Infant Sleep Questionnaire. Per-IQR early-life exposure was associated with reduced nighttime sleep duration: PM_1 β = −0.18 (95 % CI −0.32, −0.04), $PM_{2.5}$ −0.23 (−0.38, −0.08), PM_{10} −0.21 (−0.35, −0.07) hours. Quantile g-computation on the constituent mixture reproduced the decrement with SO_4^{2-} **as the dominant contributor**, and pregnancy-window findings were consistent. Outcome is parent-reported, which biases toward non-differential misclassification but not necessarily away from confounding by household environment; the three particle sizes are collinear at this resolution, so the size ordering is suggestive rather than discriminating. Sulfate as the driver is consistent with the secondary-inorganic signal this watch carried on 26 August from an unrelated endpoint.

doi: 10.3390/toxics14080653

Respiratory & allergic**9 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (9 OF 9 RECORDS IN THIS SUBTOPIC)

Aulia Anwar et al. 2026 Exp Oncol*Smoking, environmental exposures, and chronic lung disease as key risk factors for lung cancer in North Sumatra*

Hospital-based case-control across three referral hospitals in North Sumatra, August–November 2024: **240 histologically confirmed lung cancers and 240 controls** without malignancy or chronic respiratory disease, matched on sex and residential area with age adjusted in the model. Independent predictors: **unhealthy indoor environment (poor ventilation, household pollutants) OR 2.53 (95 % CI 1.45–4.40) and occupational exposure OR 1.91 (1.13–3.25)**; tuberculosis and COPD were more prevalent among cases. **Three weaknesses threaten the inference and should be stated first.** The exposure is a self-reported composite category with no measured particulate concentration, so it is subject to recall bias in a design where cases know their diagnosis. Controls were selected to be free of chronic respiratory disease, which is a collider on the pathway from household smoke to both COPD and lung cancer and biases the household-environment OR upward. And smoking differed between groups at $p = 0.023$ yet does not appear among the adjusted independent predictors — the deposited abstract does not explain that discrepancy. Treat 2.53 as an upper bound on a poorly specified exposure, not as a household-PM effect estimate.

doi: 10.15407/exp-oncology.2026.02.152

Zhongguo Fei Ai Za Zhi 2026 Zhongguo Fei Ai Za Zhi*China Wumeng Mountain lung cancer: evidence chain, clinical and molecular characteristics of regional lung cancer related to coal pollution in the Yunnan–Guizhou Plateau*

Narrative synthesis assembling an exposure → component → biological-effect → clinical-molecular → prevention chain for household smoky-coal lung cancer in Xuanwei/Fuyuan, searching PubMed, Web of Science, Embase, CNKI and Wanfang to May 2026. Carcinogenic components named: **PAHs and methylated PAHs, $PM_{2.5}$ /black carbon, inorganic minerals and metals, and carbonyls**, as a mixture. The clinical phenotype is distinctive — predominantly **never-smoking women**, high adenocarcinoma fraction, young onset, early metastasis, multifocal disease and family aggregation. So is the molecular one: total EGFR mutation is high but classical 19del and L858R are *under-represented*, while **G719X, S768I, KRAS G12C and ALK/ROS1 fusions are elevated** relative to non-Wumeng lung cancer; ARHGEF5, MYO18B and MUC16 are proposed germline susceptibility genes interacting with POT1/TERT/TERF2. Sputum microbiome shifts are reported alongside. **It is a narrative review by an unnamed author group with no formal inclusion criteria or bias assessment**, drawing on the same group's prior work, so the evidence chain is an argument rather than a synthesis. Its value is as the clearest statement in this corpus of a testable source-composition-to-mutation-spectrum hypothesis.

doi: 10.3779/j.issn.1009-3419.2026.102.16

Li et al. 2026 Int J Chron Obstruct Pulmon Dis*Air pollution and COPD hospitalisation in Kashi, Xinjiang*

Time-stratified case-crossover on **68,482 COPD patients** in Kashi, 2012–2022, with daily pollutant concentrations matched to residential address rather than to a single city monitor. Strongest PM association at **lag day 4**: odds ratios **1.015 (1.002–1.032) and 1.008 (1.002–1.016)** for the two PM fractions, with NO_2 and CO both at 1.034. Young patients, those with prior severe exacerbations, farmers, smokers and cool-season admissions were the susceptible strata. The honest part is the sensitivity analysis: **excluding COVID-19-period hospitalisations left the PM and NO_2 associations barely significant and CO non-significant**, which means a material share of the headline estimate rides on pandemic-era admission patterns. The publisher's indexing stripped every subscript from the abstract, so which PM fraction is which must be read from listing order. Kashi sits in a severe dust-storm corridor, making this one of the few large COPD series in a mineral-dust-dominated aerosol.

doi: 10.2147/copd.s617521

Morvan et al. 2026 Int J Epidemiol*Causal effect of $PM_{2.5}$ on COVID-19 hospitalisation and death, conditioned on comorbidity*

Instrumental-variable analysis over **more than 3.5 million weekly municipality observations** covering all of France, 2020–2021, with COVID-19 hospitalisations and deaths disaggregated by age, sex and municipality and linked to records of pre-existing conditions. **Wind direction instruments local $PM_{2.5}$** in a two-stage least squares design. Among individuals *without* pre-existing conditions there was **no significant effect** on either outcome. Conditioning on comorbidity, a $10 \mu g/m^3$ increase was associated with **+1.5 COVID-19 deaths per 100,000 (95 % CI 0.8–2.2)**, concentrated in respiratory (**0.9, 0.4–1.4**) and cardiovascular (**0.6, 0.1–1.1**) subgroups, and driven by the oldest age strata. Wind direction satisfies exclusion only if it affects COVID outcomes solely through $PM_{2.5}$ — doubtful during a period when weather drove both mobility and indoor crowding — and the estimates are ecological rate differences, not individual risks. Still the strongest identification strategy this literature has produced.

doi: 10.1093/ije/dyag152

Silva et al. 2026 Front Sleep*Shared mechanisms linking PM_{2.5} exposure, obstructive sleep apnea and obesity*

Narrative review of the three-way relationship between chronic PM_{2.5} exposure, obstructive sleep apnea and obesity, proposing systemic inflammation, oxidative stress, neuroendocrine dysregulation and gut microbiota alteration as shared pathways, and noting that intermittent hypoxia and sleep fragmentation feed back into weight gain. The mechanistic overlap is plausible and the review is explicit that it is hypothesis-organising, but **no effect estimates, no study inventory and no appraisal of confounding by adiposity** appear — and adiposity confounds both the exposure (residential location) and the outcome. Read alongside the CHARLS sleep-duration result carried on 28 August, the practical gap is identical in both: no objective sleep measurement anywhere in this literature.

doi: 10.3389/frsle.2026.1918405

Wang & Li 2026 Front Public Health*Air pollution and atopic dermatitis*

Narrative review of PubMed, Web of Science and CNKI, April 2021 to December 2025, restricted to peer-reviewed work with impact factor ≥ 3.0 — a screening criterion that is a proxy for nothing and silently excludes most dermatology and most regional epidemiology. Covers epidemiology, skin-barrier and microbiome mechanisms, climatic and geographical disparity, and prevention. No quantitative synthesis and no effect estimates. Carried at tier C as a citation anchor for the cutaneous endpoint, which is otherwise absent from this watch.

doi: 10.3389/fpubh.2026.1854212

Yang et al. 2026a GeoHealth*Spatiotemporal risk of environmental drivers of respiratory disease, Jiangsu*

Distributed lag non-linear models coupled to geographically and temporally weighted regression on respiratory hospitalisations across Jiangsu Province, 2020–2023, a population containing over 27 million children under 15 and adults over 65. The DLNM handles the lag structure and the GTWR lets the exposure–response coefficient vary in space and time rather than assuming one province-wide slope — which is the right question for a province with this much industrial heterogeneity. The 2020–2023 window overlaps pandemic-era mobility and healthcare-seeking disruption; the abstract does not say how that is handled, and it is the main threat to the temporal component.

doi: 10.1029/2026gh001881

Zhongguo Fei Ai Za Zhi 2026 Zhongguo Fei Ai Za Zhi*China Wumeng Mountain lung cancer: evidence chain, clinical and molecular characteristics of regional lung cancer related to coal pollution in the Yunnan–Guizhou Plateau*

Narrative synthesis assembling an exposure → component → biological-effect → clinical-molecular → prevention chain for household smoky-coal lung cancer in Xuanwei/Fuyuan, searching PubMed, Web of Science, Embase, CNKI and Wanfang to May 2026. Carcinogenic components named: **PAHs and methylated PAHs, PM_{2.5}/black carbon, inorganic minerals and metals, and carbonyls**, as a mixture. The clinical phenotype is distinctive — predominantly **never-smoking women**, high adenocarcinoma fraction, young onset, early metastasis, multifocal disease and family aggregation. So is the molecular one: total EGFR mutation is high but classical 19del and L858R are *under-represented*, while **G719X, S768I, KRAS G12C and ALK/ROS1 fusions are elevated** relative to non-Wumeng lung cancer; ARHGEF5, MYO18B and MUC16 are proposed germline susceptibility genes interacting with POT1/TERT/TERF2. Sputum microbiome shifts are reported alongside. **It is a narrative review by an unnamed author group with no formal inclusion criteria or bias assessment**, drawing on the same group's prior work, so the evidence chain is an argument rather than a synthesis. Its value is as the clearest statement in this corpus of a testable source-composition-to-mutation-spectrum hypothesis.

doi: 10.3779/j.issn.1009-3419.2026.102.16

de Souza et al. 2026 Sci Total Environ*Wildfire smoke, PM_{2.5} and respiratory morbidity: spatial mapping and time series*

1,467,172 respiratory hospital admissions in São Paulo State, 2018–2025 (DATASUS), against **20,446 INPE-recorded wildfire events** and daily CETESB PM_{2.5}, temperature and humidity, analysed by GAM on weekly aggregates with adjustment for meteorology, calendar month and a nonlinear trend. Children 0–10 and males carried the greatest burden; population-standardised rates were highest in the north and north-west, coinciding with fire hotspots, while absolute counts concentrated in the metropolitan area — a textbook illustration of why per-capita and count maps disagree. **Each 10 µg/m³ weekly PM_{2.5} was associated with a 3.4% increase in admission rates (RR 1.034, 95% CI 0.889–1.204, $p = 0.054$)**, strongest at zero lag. Fire counts and PM_{2.5} correlated at $\rho = 0.732$. The GAM explains 82.8% of deviance, but weekly aggregation of an acute exposure throws away most of the contrast and is the likeliest reason the interval spans the null.

doi: 10.1016/j.scitotenv.2026.182227

Reproductive & developmental**3 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (3 OF 3 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026g Toxics*Prenatal size-fractionated PM and fetal growth trajectories, mediated by maternal glucose*

616 mother–fetus pairs in Guangzhou with repeated mid-to-late-pregnancy ultrasound (HC, AC, FL, BPD as z-scores; EFW trajectories classed average, slow or rapid) and residential PM₁, PM_{2.5}, PM₁₀ from 1 km spatiotemporal models. Each 5 µg/m³ increase raised the odds of a *rapid* EFW trajectory versus average: **PM₁ OR 1.56 (1.03–2.37), PM_{2.5} 1.44 (1.10–1.88), PM₁₀ 1.28 (1.05–1.56)** — monotone in the inverse of particle size, as a deposition argument would predict. BPD, HC and AC z-scores moved in the same direction. **Maternal fasting glucose and insulin resistance mediated 5.0–29.4% of the association**. The mediation range is wide enough to be uninformative at its lower end, mid-pregnancy glucose is a single measurement, and *accelerated* fetal growth is a less established adverse outcome than restriction, so the clinical framing is the weakest part. The size gradient is the citable result.

doi: 10.3390/toxics14080709

Kessler et al. 2026 Environ Health (Wash)*Prenatal traffic-related air pollution, dietary PUFA intake and birth weight*

Atlanta Maternal–Child Cohort, **297 pregnant African American women**, first-trimester traffic-related CO, PM_{2.5} and NO_x crossed with dietary polyunsaturated fatty acid intake in a joint mixture model. At mean α -linolenic acid intake, a per-IQR PM increase was associated with a **0.08-unit decrease (0.02–0.18) in birth-weight-for-gestational-age z-score**; at twice mean ALA intake the same increase was associated with a **1.12-unit increase (0.19–2.05)**, $p_{\text{int}} = 0.04$, with an almost identical pattern for linoleic acid. The authors read this as PUFA mitigation. A cleaner reading is that a 14-fold sign-reversing interaction estimated on 297 women in the sparse upper tail of a food-frequency distribution is the signature of sparse-data bias. Worth tracking because the mechanistic story is plausible and the result will be cited widely.

doi: 10.1021/envhealth.5c00490

Wang et al. 2026 Environ Health (Wash)

Ultrafine particles, hypertensive disorders of pregnancy, and homocysteine mediation

Shanghai birth cohort, **7,569 pregnancies**, 2017–2018, with individual UFP exposure from a high-spatiotemporal-resolution prediction model at residential address and serum homocysteine, folate and B₁₂ measured in early pregnancy. Poisson regression with robust error variance gave per-IQR RR **1.15–1.23 for hypertensive disorders of pregnancy and 1.29–1.41 for pre-eclampsia** across the prepregnancy, first- and second-trimester windows. Homocysteine was associated with pre-eclampsia (RR **1.49, 1.04–2.13**) and mediated 8.87% and 2.51% of the prepregnancy and first-trimester associations. Stratification showed the UFP-PE association only in the low-B₁₂ group. Two cautions: the exposure is modelled, not measured, so UFP-PM_{2.5} collinearity is not resolvable here; and mediated fractions this small are within the range that measurement error in the mediator can manufacture.

doi: 10.1021/envhealth.5c00760

Mechanistic toxicology

12 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (12 OF 12 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026j Part Fibre Toxicol

Multi-omics integration reveals human placental molecular signatures associated with microplastics exposure

Placental microplastic burden quantified by **pyrolysis-gas chromatography–mass spectrometry**, samples stratified into high- and low-burden groups, then transcriptomic, proteomic and metabolomic profiling integrated through the DIABLO framework with network analysis for regulatory hubs. High-burden placentae show multi-layer molecular perturbation centred on immune dysregulation. The design's strength is that MP burden is treated as a *graded internal dose* rather than a detection/non-detection flag, which separates it from most human-tissue microplastics work. Its limits are the usual ones for cross-sectional human tissue: burden is measured once at delivery, exposure timing is unknown, the omics contrasts are correlational, no sample size appears in the abstract, and Py-GC-MS quantifies polymer *mass* without particle size or count — so the dose metric is mass, not number or surface area.

doi: 10.1186/s12989-026-00697-2

Forest 2026 Sci Total Environ

Molecular mechanisms of sand and dust storm toxicity

Review of in vitro and in vivo evidence on storm-derived PM₁₀/PM_{2.5}, whose composition — quartz and clay minerals plus adsorbed bacteria, viruses and heavy metals acquired in transport — differs enough from urban PM that the author argues current regulation, which treats storm episodes as generic PM, misses hazard-specific risk. Consistent findings are cytotoxicity, oxidative stress, pro-inflammatory response and exacerbation of bacterial and viral infection. The stated gaps are the substantive content: **most in vivo work uses Asian sand dust while Saharan dust dominates global emissions**, and non-respiratory target organs are barely studied. Directly relevant to any network operating in a dust-influenced region, where the mass a sensor reports and the hazard it implies decouple.

doi: 10.1016/j.scitotenv.2026.182233

Li & Li 2026 Zhongguo Fei Ai Za Zhi

Research advances in the pulmonary carcinogenic effects of atmospheric fine particulate matter mediated by GPER

Narrative review positioning the G protein-coupled estrogen receptor as the molecular hub linking PM_{2.5} to lung carcinogenesis. The argument: environmental estrogen-like organics and metal ions carried on PM_{2.5} activate GPER, which is overexpressed in lung tumour tissue, and the activated receptor drives **MAPK/ERK and PI3K/AKT** signalling into inflammation, oxidative stress and ferroptosis, with downstream proliferation, EMT and tumour-microenvironment remodelling. GPER inhibition is proposed as a target. **The weaknesses are structural to the format:** no search protocol, no risk-of-bias assessment, no quantitative synthesis, and a receptor-centric frame that cannot by construction weigh GPER against the AhR and oxidative-stress pathways it does not review. Read it as a hypothesis map, not as evidence of pathway dominance. The one genuinely actionable implication for exposure science is that it assigns the causal role to specific *components* — estrogenic organics and transition metals — rather than to mass, which is another argument for composition-resolved rather than gravimetric-equivalent monitoring.

doi: 10.3779/j.issn.1009-3419.2026.101.13

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doi: 10.3779/j.issn.1009-3419.2026.101.13

Liu et al. 2026 Environ Health (Wash)

PM_{2.5}-induced pulmonary vascular remodelling via MSC-mediated endothelial senescence

Real-ambient PM_{2.5} exposure in mice plus a pulmonary arterial endothelial cell model. Exposure raised pulmonary artery pressure and arterial wall thickness in vivo and induced senescence and mitochondrial dysfunction in mouse pulmonary arterial endothelial cells. Bioinformatic screening nominated the transcription factor MSC as the regulator; **MSC overexpression rescued mitochondrial-dysfunction-associated senescence**, and MSC was shown to act through the downstream target NDUF3, a complex I subunit — which makes the mitochondrial phenotype mechanistically coherent rather than merely correlative. **Real-ambient rather than instilled exposure** keeps the dosimetry defensible. No concentration, duration or particle characterisation appears in the abstract, so comparability to other real-ambient systems cannot be judged.

doi: 10.1021/envhealth.5c00595

Moonwiriya et al. 2026 J Vis Exp*A STAT6 reporter assay for screening toxicants affecting allergic airway inflammation*

A methods paper establishing a stable STAT6-responsive luciferase reporter in **16HBE14o- human bronchial epithelial cells** by lentiviral transduction, with puromycin selection, MOI optimisation and monoclonal isolation by limiting dilution. Responsiveness was validated against IL-4/IL-13, pathway specificity confirmed with selective STAT6, STAT5 and STAT3 inhibitors, and reproducibility quantified by **Z'-factor**. Applied to individual air-pollution constituents, **benzo[b]fluoranthene and particulate matter both significantly increased STAT6 reporter activity**. No dose-response, no PM characterisation (source, size, mass loading) and no Z' value are given in the abstract, so the assay's screening sensitivity cannot be assessed from it. The value is infrastructural: allergic-airway signalling is a plausible route for PM-associated asthma incidence and there has been no scalable prioritisation assay for it.

doi: 10.3791/71694

Nikaido-Landry et al. 2026 Philos Trans R Soc Lond B Biol Sci*EGFR dysregulation and climate change: pollution-promoted lung cancer*

Discussion-meeting review connecting rising anthropogenic fine PM to lung cancer in never-smokers via EGFR signalling. The mechanistic core is the promotion model: **fine PM promotes malignant transformation in cells that already carry an EGFR mutation**, so the relevant population is people with oncogenic mutations in histologically normal lung tissue rather than people accumulating new mutations from exposure. That reframing matters — it predicts a susceptible subgroup rather than a population-wide dose-response, and it implies precision-prevention strategies targeted by genotype rather than universal exposure reduction. A narrative review with no systematic search and no new data; the promotion model rests principally on one line of murine and cohort work, and the “climate change” framing in the title is rhetorical rather than analytic. Tier B for the mechanistic argument.

doi: 10.1098/rstb.2024.0508

Pathak et al. 2026 medRxiv (preprint)*Airborne PM and mitochondrial DNA copy number: systematic review and meta-analysis*

PRISMA, PROSPERO-registered, JBI risk-of-bias, random-effects pooling; 1,224 records screened to 24 studies and **13,092 participants**, 12 reporting percentage change and 12 absolute values. **Percentage mtDNA-CN fell** (-4.90 ; **95% CI** -7.97 to -1.82 ; $p = 0.002$) **while absolute mtDNA-CN rose** ($+0.55$; **0.05–1.04**; $p = 0.030$). The two directions are not reconcilable as biology; they are an artefact of pooling across incompatible normalisation schemes, and the honest reading is that the underlying literature has not standardised its outcome. Useful as evidence that mtDNA-CN is *not yet* deployable as a PM exposure biomarker, which is the opposite of how it is usually cited.

doi: 10.64898/2026.08.20.26360559

Qin et al. 2026 Biofactors*AlCl₃ aerosol exposure is associated with pulmonary lactate accumulation, neutrophil dysfunction, and fibrosis-like remodeling*

Repeated AlCl₃ aerosol exposure in rats, with and without the MCT1 inhibitor AZD3965, plus bone-marrow-derived neutrophils treated with sodium L-lactate, plus a supportive human cohort of aluminium-exposure-related ILD, non-aluminium ILD and healthy controls characterised by quantitative CT, spirometry and blood transcriptomics. Exposure produced pulmonary aluminium deposition, **lactate accumulation**, oxidative stress, DNA damage response signalling and collagen-rich fibrosis-like remodelling; in vitro, high lactate impaired neutrophil chemotaxis, oxidative burst, mitochondrial membrane potential, calcium signalling and NET responses. AZD3965 partially reversed both. Al-ILD patients showed greater CT parenchymal abnormality and lower FVC and DLCO than non-Al-ILD, with innate-immune, cytokine, ECM and NET pathways enriched in blood. **No exposure concentration, particle size distribution or duration appears in the deposited abstract**, so the animal arm cannot be placed on any dose axis, and the human arm is cross-sectional with no exposure measurement at all — it supports the mechanism narratively and cannot test it. Relevant here as occupational-aerosol toxicology and as a reminder that metal-bearing particulate has organ-level effects that mass concentration does not index.

doi: 10.1002/biof.70146

Qin et al. 2026b Environ Health (Wash)*Engineered protein coronas: ApoE enhances particulate transport across the blood-brain barrier*

Gold nanoparticles functionalised with thiolated DNA aptamers serve as a **compositionally controlled PM surrogate**, allowing reproducible assembly of coronas from three physiologically relevant proteins — human serum albumin, apolipoprotein E and complement C3b — and isolating corona identity from particle identity. In a bEnd.3 Transwell BBB model, **ApoE-enriched coronas nearly doubled apparent permeability and increased 24 h transport 1.6-fold** over bare particles, through an energy-dependent, predominantly clathrin-mediated route. ApoE precoronation also raised endothelial penetration of polystyrene particles and **increased the endothelial cytotoxicity of real industrial-source PM samples**, which is the step that connects the model to ambient aerosol. The surrogate is the limitation: gold nanoparticles share neither the surface chemistry nor the morphology of combustion PM, and a Transwell monolayer is not a neurovascular unit. Mechanistically the cleanest account yet of why particle *surface*, not particle mass, governs neural access.

doi: 10.1021/envhealth.5c00803

TARGET II Consortium 2026 bioRxiv (preprint)*Lifecourse sex-specific molecular response to early-life toxicant exposure*

Consortium-scale design: mice exposed from two weeks preconception through gestation and lactation to arsenic, lead, tributyltin, BPA, DEHP, TCDD or PM_{2.5}, with **1,025 omics datasets** generated from liver across three life stages. The headline is a significant sex-biased transcriptomic and epigenomic response to a common exposure. For this watch the value is comparative — PM_{2.5} is run alongside six reference toxicants under one protocol, which is the only way to judge whether a PM molecular signature is distinctive or generic. Liver-only, and mouse inhalation dosimetry does not scale simply to humans. Unreviewed.

doi: 10.64898/2026.08.20.746014

Van et al. 2026 In Vivo*MASLD exacerbates chronic low-dose PM_{2.5}-induced lung injury, inflammation and fibrosis*

C57BL/6J mice on a Western or normal diet for 28 weeks with concurrent intratracheal diesel particulate matter at **0.5 mg/kg three times weekly** (stated as environmentally relevant to $\sim 50 \mu\text{g}/\text{m}^3$), assessed by histology, Masson's trichrome, Luminex multiplex on bronchoalveolar lavage and lung-tissue ELISA/qPCR. The Western diet induced the MASLD phenotype; PM_{2.5} did not worsen the hepatic phenotype, but **the combination markedly exacerbated pulmonary injury and fibrosis relative to PM_{2.5} alone**. The dose-equivalence claim is the weak point: bolus intratracheal instillation of diesel PM delivers a deposition profile nothing like chronic inhalation at $50 \mu\text{g}/\text{m}^3$, so the “environmentally relevant” framing should not be taken at face value. The interaction direction is the transferable result.

doi: 10.21873/in vivo.14428

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (7 OF 7 RECORDS IN THIS SUBTOPIC)

Li & Zheng 2026 Toxics*Size-resolved PM₁₀-bound PFAS: respiratory deposition and oxidative stress in recycling workers*

Panel study at a southern-Chinese waste recycling plant: **15 size-resolved PM samples and 280 repeated first-void urines from 20 workers over 45 consecutive days**, with linear mixed models linking size-resolved PM-bound PFAS to urinary PFAS and to 8-OHdG and MDA. Short-chain PFAS predominated in both matrices; urinary ICCs were fair to excellent (>0.4). **PFAS in PM_{2.1-10} correlated more strongly with internal dose than PFAS in PM_{2.1}** — the opposite of the fine-fraction-dominates prior, and consistent with mucociliary clearance and gastrointestinal uptake of coarse-fraction deposits. **Each unit increase in ln average daily intake was associated with 10.2–12.6 % higher 8-OHdG (PM_{4.7-5.8}, PM_{9.0-10}) and 8.0–14.6 % higher MDA (PM_{5.8-9.0})**. Twenty workers, one plant, and the authors call the biomarker associations suggestive; with this many size bins the multiplicity is substantial and no correction is described. The size-resolved design is nonetheless the right one and is almost never done.

doi: 10.3390/toxics14080682

Ospalek et al. 2026 Atmosphere*Air quality assessment and potential inhalation exposure across equestrian facilities*

Nine measurements at 1.5 m in a stable, an indoor riding arena and an outdoor riding arena at a Slovak university equestrian centre, covering PM₁, PM_{2.5}, PM₁₀, respirable and total dust plus gaseous pollutants, with four exposure scenarios combining 18–20 h stabled with 4–6 h ridden. **The outdoor arena carried the highest concentrations of every PM fraction and the indoor arena the lowest ($p < 0.05$)** — the opposite of the expected enclosure effect, and the paper's most interesting result; ammonia was higher in the stable, as expected. Summer measurements only, one facility, nine measurements total, and the exposure scenarios are constructed rather than observed, so the inhaled-dose estimates are illustrative. Relevant here as a template for microenvironment-resolved dose reconstruction from sparse fixed-point sampling.

doi: 10.3390/atmos17090840

Perna et al. 2026 Life (Basel)*Indoor air pollution as a determinant of airport occupational exposure*

Narrative review arguing that airport occupational exposure assessment has been anchored on aircraft engine emissions and has under-weighted **indoor air quality driven by indoor–outdoor exchange and ventilation**. Ultrafine particles, black carbon, PAHs and trace metals are identified as the constituents driving oxidative stress, inflammation and xenobiotic pathway activation, with effects framed beyond the respiratory system. The operationally useful recommendation is to monitor indoor microclimatic parameters — temperature, pressure, humidity — **as event detectors for indoor air-quality deterioration**, which is a cheap addition to any fixed deployment. No systematic search, no quantitative synthesis, no exposure or effect estimates; the closing suggestion that AI-based exposure modelling is promising is asserted with the authors' own caveat that it lacks validation. Tier C: useful as a framing document for airport deployments, not as evidence.

doi: 10.3390/life16081242

Ramirez et al. 2026 Curr Allergy Asthma Rep*Airway diseases and occupational exposures in poultry industry workers: a comprehensive review*

Review of occupational asthma and chronic obstructive lung disease in poultry workers, attributing risk to the barn's mixed aerosol — feather and feed dust, mites, moulds, ammonia and bacterial endotoxin — acting through airway inflammation and hypersensitivity, with a symptom spectrum from wheeze and chest tightness to chronic cough and dyspnoea. Emphasis falls on early identification through occupational history-taking and on barn control practices. **No prevalence figures, no exposure–response estimates and no search or inclusion strategy** are reported, so this is a clinical orientation piece rather than an evidence synthesis. Its relevance here is the exposure architecture: an occupational aerosol whose health-relevant fraction is defined by biological composition, which a mass-only monitor cannot resolve at all.

doi: 10.1007/s11882-026-01289-y

Sadki et al. 2026 Cad Saude Publica*Real-time formaldehyde exposure and acute symptoms in healthcare workers*

Cross-sectional study of **180 healthcare workers across 12 hospital departments** in Morocco, with airborne formaldehyde measured over full eight-hour shifts by calibrated semiconductor dosimeter alongside a structured symptom questionnaire. **Every mean concentration was below the Moroccan occupational limit of 0.25 mg/m³ (0.048–0.196 mg/m³, 19–78 % of the limit)**, yet highly exposed workers reported markedly more eye irritation (72.5%), headache (63%) and respiratory discomfort (56.5%). **Prolonged exposure (≥ 8 h/day, ≥ 3 days/week) independently predicted multiple symptoms (AOR 3.2, 95 % CI 1.7–5.9)**, as did inadequate ventilation (2.4) and inconsistent PPE use (1.9); years of exposure correlated with symptom count ($\rho = 0.41$). **The exposure is gaseous, not particulate**, so the AORs are excluded from this issue's forest plot. Carried because the deployment pattern — low-cost real-time dosimetry as routine occupational surveillance in a resource-limited setting — is exactly the use case for distributed sensing, and because symptom self-report against measured exposure is open to reporting bias in the obvious direction.

doi: 10.1590/0102-311XEN191325

Siemens et al. 2026 Indoor Air*A SynAir-G position statement for the improvement of indoor air quality for children*

Position statement from the EU-funded SynAir-G project (18 partners, 11 countries) within the seven-project IDEAL Cluster, arguing that indoor air has been under-regulated relative to outdoor air despite most exposure occurring indoors, and that the policy-relevant unit is the **synergistic interaction between chemical and biological pollutants** rather than any single-pollutant concentration. Schools and children are the target setting. As a consensus document it presents no new measurements, no systematic evidence appraisal and no quantified synergy, so the central claim is asserted rather than demonstrated. The instrumentation implication is nonetheless concrete: a synergy framing requires co-located chemical and bioaerosol sensing at classroom scale, which no current low-cost platform delivers.

doi: 10.1155/ina/9690417

Yang et al. 2026b Environ Pollut*Bioaerosols at oil exploitation sites and mask filtration efficiency*

Seasonal sampling in a Beijing–Tianjin–Hebei oil city carrying both heavy PM and VOC loads, characterising culturable and total bioaerosol distribution, sources and inhalation risk, plus mask filtration efficiency. **Autumn levels significantly exceeded summer while microbial activity moved the other way** — a dissociation between count and viability that matters for anyone interpreting a fluorescence channel as a proxy for biological hazard. Culture-based enumeration undercounts viable-but-non-culturable organisms by a large and site-dependent factor, which is the standing limitation of this entire measurement class. Tier C.

doi: 10.1016/j.envpol.2026.129020

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (13 OF 13 RECORDS IN THIS SUBTOPIC)

Acat et al. 2026 Atmosphere*National trends in ambient pollutants and cause-specific mortality, Türkiye 2018–2024*

Seven annual national observations for PM₁₀, SO₂, CO, NO₂ and O₃, each the unweighted mean of eligible station annual means from official bulletins, against crude national mortality rates, tested by OLS with Mann–Kendall, Sen's slope and a strict common-station sensitivity analysis. **PM₁₀ fell from 50.4 to 41.0 µg/m³** and SO₂ from 13.2 to 8.5, both significant. The all-station NO₂ indicator rose from 29.2 to 34.4 but **the trend vanished under common-station restriction** (slope -0.021 , $p = 0.950$) — a clean demonstration that a national mean over a changing monitoring network measures the network as much as the air. **PM_{2.5} was excluded outright** for insufficient reporting comparability, which is itself the story for a country whose modelled PM_{2.5} burden the AirQ+ record above puts at 68,440 deaths. Only neoplasm mortality declined significantly. Crude national rates against unweighted station means; no causal reading is possible and none is claimed.

doi: 10.3390/atmos17090832

Anon et al. 2026d Prev Med Rep*Lifetime risk of respiratory-system cancers across 185 countries*

GLOBOCAN 2022 combined with UN life tables under an adjusted-for-multiple-primaries framework. **Global lifetime risk of developing lung cancer 4.92 % and of dying from it 4.08 %**, against 0.33 % and 0.20 % for laryngeal cancer, with a steep gradient from **0.34 % in low-HDI to 5.45 % in very-high-HDI countries**. All four pollutants were positively associated with lifetime risk in univariable linear regression, PM_{2.5} strongest. Two hard limits: the HDI gradient in lung cancer incidence is dominated by historical smoking prevalence and by diagnostic capacity, and country-level pollution from **OpenWeather** is a consumer weather API rather than a validated exposure product. Univariable ecological regression across 185 countries cannot separate PM from GDP, smoking or registry completeness. Tier B for the burden estimates; the pollution association should not be cited as evidence of effect.

doi: 10.1016/j.pmedr.2026.103602

Anon et al. 2026e Res Sq (preprint)*A Harmonization Index for industrial PM emission regulations*

A Regulatory Coding Framework converting legally binding documents into standardized scores across **21 indicators in three dimensions** — regulatory stringency, institutional design and enforcement capacity — aggregated into a composite Harmonization Index, applied to the EU, US, China and Uzbekistan as exemplars of technology-based, source-category-based, hybrid and dispersion-based regulatory architectures. The problem is genuine: industrial PM limits are not comparable across jurisdictions because they are not the same kind of object. The method is documentary coding of what is written, which measures statutory text and not emissions; the enforcement-capacity dimension is the one that would need external validation and is the one hardest to score from documents. Unreviewed, four jurisdictions, tier C.

doi: 10.21203/rs.3.rs-10491363/v1

Cliff et al. 2026 GeoHealth*How do climatic factors directly influence the incidence and risk of meningococcal meningitis across the African meningitis belt?*

Narrative review over the 26-country African meningitis belt, where outbreaks cluster strongly in the dry season, examining dust, wind speed, temperature, rainfall and land cover as candidate drivers. **Atmospheric dust and wind speed carry the strongest statistical association and the greatest predictive probability** of the variables examined, with mucosal damage from inhaled mineral dust the standing mechanistic hypothesis. The review's own central caveat is the right one: these climatic variables are severely multicollinear within the dry season, so attributing incidence to dust rather than to temperature or humidity is not identified by the observational record. Carried here because it is one of the few live health endpoints where the *coarse* mineral fraction, not PM_{2.5}, is the hypothesised agent — and where satellite dust products substitute for a ground network that does not exist.

doi: 10.1029/2025gh001749

Nilsson et al. 2026 Environ Epidemiol*Urban environment, health and equity: symptoms and annoyances related to air pollution, traffic noise and greenness*

46,282 adults aged 18–84 in Stockholm County responding to the 2023 Swedish National Environmental Health Survey, with geocoded residential PM_{2.5}, PM₁₀, NO₂, road/rail/aircraft noise and 300 m NDVI, dichotomised and stratified by socioeconomic status and building type. Apartment residents are more exposed to pollution and noise and have less greenness; SES-stratified patterns are exposure-specific with no consistent overall gradient. The strong associations are with *perception* — NO₂ with poor perceived outdoor air quality (**OR 3.22; 95% CI 2.75–3.79**), road noise with high annoyance (OR 4.59; 4.02–5.26). Both exposure and outcome are cross-sectional and the outcome is self-reported, so these are perception–exposure concordance estimates, not health effects, and the odds ratios should not be read as such.

doi: 10.1097/EE9.0000000000000516

Özmen et al. 2026 Toxics*Long-term PM_{2.5} and premature mortality across Türkiye*

Province-level cross-section of **all 81 Turkish provinces** for 2022, combining annual mean PM_{2.5} with WHO AirQ+ attributable-death estimates for adults 30+ at a 5 µg/m³ counterfactual. **Population-weighted mean PM_{2.5} was 27.0 µg/m³ — 5.4× the WHO guideline — and all 81 provinces exceeded it**. Bivariate OLS explained 41 % of between-province variation in attributable mortality (**slope 3.23 additional deaths per 100,000 per 1 µg/m³, 95 % CI 2.37–4.10**); a log-log model gave an elasticity of **0.80 (0.65–0.95)**. The attributable fraction of natural-cause mortality rose from **8.8 % in the lowest exposure quartile to 24.6 % in the highest**, totalling **68,440 premature deaths, 14.2 % of natural-cause mortality**. The authors are explicit about the circularity and it is worth repeating: attributable deaths are *generated* by AirQ+ from a fixed concentration–response function, so regressing them on concentration recovers the assumed function, not independent evidence. The burden magnitude stands; the gradient is arithmetic.

doi: 10.3390/toxics14080728

Sharma et al. 2026b Health Promot J Austr*Geospatial linkage of environmental exposures, socioeconomic profile and health outcomes among women of reproductive age, Victoria*

Postcode-level linkage of 2021 Australian Census sociodemographic and health variables to PM_{2.5}, NO₂, CO, SO₂, formaldehyde, ozone, land surface temperature and greenness, with Moran's I, Wagstaff concentration indices and geospatial regression. Non-communicable disease burden in women of reproductive age is strongly socioeconomically patterned (**323.0 vs. 227.3 per 1000** in most- versus least-disadvantaged postcodes; steepest for diabetes, Wagstaff -0.16), **but environmental exposures show only weak associations with outcomes** despite clear spatial clustering. Inner Melbourne has lower NCD burden despite elevated NO₂ and PM_{2.5}. This is ecological, cross-sectional and confounded by the same SES gradient it measures; the null environmental finding is uninformative rather than reassuring.

doi: 10.1002/hpja.70241

Sirikul et al. 2026 J Clim Chang Health

Knowledge, attitudes and practices of caregivers regarding PM_{2.5} exposure among children in northern Thailand

Cross-sectional KAP survey of caregivers recruited at Chiang Mai Child Development Centres. Knowledge is moderate (10.44 ± 3.80 of 22), attitudes positive (30.74 ± 3.57 of 40), practices inconsistent (17.33 ± 5.22 of 32). The specific gaps are the useful part: **97.0% recognise the health risk to children, but only 41.8% know surgical masks are inadequate for PM_{2.5} and only 9.7% can correctly interpret an AQI value**; air purifier ownership is 15.3% and monitoring-app use 28.2%, both with a clear education and income gradient. The abstract does not report the sample size, response rate or instrument validation, and centre-based recruitment excludes non-attending households, so absolute prevalences are not generalisable. The AQI-literacy figure is the one that should worry anyone designing a public-facing sensor dashboard.

doi: 10.1016/j.joclim.2026.100734

Sommar et al. 2026 Int J Environ Res Public Health

Health economics of PM_{2.5} from residential wood combustion, northern Sweden

Health impact assessment for **280,742 residents of Västerbotten County** using high-resolution 2023 dispersion-modelled PM_{2.5} from small-scale residential heating, population-weighted to home-address coordinates, with the VALESOR framework and a near-source relative risk of **1.26 per 10 µg/m³** for natural-cause mortality. **Mean RWC-attributable exposure was only 0.16 µg/m³** (IQR 0.10–0.21), yielding **30 years of life lost annually** plus small morbidity counts (0.3 lung cancers, 4 childhood asthma, 3 COPD, 3 IHD, 1 stroke, 2 dementia) and a total annual cost of **~EUR 7.0 million (EUR 4.6–9.4 million)**. Two assumptions carry the result: the near-source RR of 1.26 is far above the standard ~ 1.08 per 10 µg/m³ and is a deliberate wood-smoke-toxicity premium that the evidence base supports only weakly, and the population-mean of 0.16 µg/m³ conceals a highly skewed near-source distribution. Fractional case counts at this precision are not meaningful.

doi: 10.3390/ijerph23081053

Wang et al. 2026 Front Glob Womens Health

Trends, inequalities, and 2030 projections of women's tracheal, bronchial, and lung cancer attributable to air pollution, radon, and tobacco exposure

GBD 2023, female tracheal/bronchial/lung cancer 1990–2023, five inhalation-related risk factors, with SDI-gradient smoothing, concentration indices for inequality, decomposition by demographic vs epidemiological change, and projection to 2030. Smoking remains the leading contributor, **ambient particulate matter second, secondhand smoke third**. Over 1990–2023 **household solid-fuel-attributable mortality fell most sharply while particulate-matter-attributable mortality rose slightly**, with China the standout for the PM-attributable share. Along the SDI gradient, PM shows a **rising burden with diminishing inequality** — the opposite pattern to radon and tobacco — and decomposition attributes the PM increase to population growth and aging rather than to rising risk. Projected to keep rising to 2030. Standard GBD caveats bind hard here: the exposure is a modelled global surface, the attributable fraction rests on assumed counterfactual and RR functions, and the 2030 projection is an extrapolation of fitted trend, not a scenario.

doi: 10.3389/fgwh.2026.1662451

Zhai et al. 2026 Front Public Health

Global disparities in lifetime lung cancer risk: development, smoking and fine PM

GLOBOCAN 2022 with the adjusted-for-multiple-primaries method, giving **global lifetime risk of lung cancer incidence 4.91 % and mortality 4.07 %**, ranging from North America (7.30 % / 5.38 %) to Africa (0.99 % / 0.93 %) and higher in males on every continent. In country-level regression, smoking prevalence predicted both (LRI $\beta = 0.001$, $p < 0.001$), and **PM₁ was the strongest environmental predictor** (LRI $\beta = 0.40$, $p = 0.001$; LRM $\beta = 0.20$, $p = 0.006$), slightly ahead of PM_{2.5} (0.36 and 0.18). Higher national development correlated with higher lifetime risk. The ecological confounding is total: HDI, smoking history, registry completeness and diagnostic capacity all covary with PM at country level, and lung cancer incidence in high-HDI countries is dominated by historical smoking and by the ability to detect it. **PM₁ outranking PM_{2.5} is the one result worth carrying**, and even that is weak given how few countries have independent PM₁ products.

doi: 10.3389/fpubh.2026.1842037

Zhang et al. 2026g Nat Commun

Health and economic impacts of improved air quality from transitioning global shipping energy to hydrogen

Integrated emission–air quality–health framework over feasible hydrogen-transition scenarios for global shipping. Full adoption averts approximately **129,500 (95% CI 68,300–202,200) premature deaths annually**, valued at **US\$306 bn (146–488)**, with India, Japan and the United States taking the largest economic benefit. The number that matters for a PM watch is the split: **75% of the averted mortality is ozone, only 25% PM_{2.5}**, with the O₃ gains concentrated in Southern Asia and the PM_{2.5} gains in Eastern Asia. The confidence interval spans a factor of three and propagates concentration–response uncertainty only, not scenario or inventory uncertainty, so the true spread is wider than stated.

doi: 10.1038/s41467-026-76134-x

Zhu et al. 2026 Atmos Pollut Res

Short-term effectiveness of urban road cleaning for fugitive dust, from TRAKER mobile monitoring

Metadata-only deposit. TRAKER (Testing Re-entrained Aerosol Kinetic Emissions from Roads) infers road-dust emission potential from the difference between behind-tyre and background aerosol measured on a moving vehicle, which makes it one of the few mobile methods that measures an *emission* rather than a concentration. The question in the title — how long the benefit of a street-sweeping pass actually persists — is one municipalities repeatedly fund cleaning programmes without answering, and the honest prior from the existing literature is “hours to a few days, if at all”. No abstract, no site, no numbers. Carried at tier C and retained because the method class is directly relevant to mobile PM sensing.

doi: 10.1016/j.apr.2026.103190

Other clinical endpoints

8 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (8 OF 8 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026h Metabolites

Metabolic perturbation in elderly adults exposed to PM_{2.5}-bound ester compounds

258 Fuzhou residents aged 60+ with 72-hour personal PM_{2.5} sampling (UPAS V2), GC-MS speciation and LC-MS plasma metabolomics, targeting non-phthalate, non-organophosphate esters that routine speciation ignores. Mean personal PM_{2.5} was **38.06 µg/m³** with esters dominating the measured organic composition; 20 high-concentration esters were prioritised. Covariate-adjusted linear regression associated ester exposure with **steroid biosynthesis, glycolysis/gluconeogenesis, glycerophospholipid and purine/pyrimidine** pathway alterations. Personal rather than ambient sampling in an elderly cohort is the strength and is uncommon; against that, the ester annotations are putative, the design is cross-sectional with a single 72-h exposure window against a single metabolome, pathway enrichment on untargeted features has a well-known false-discovery problem, and the authors state the work is hypothesis-generating. No effect sizes are reportable from the abstract.

doi: 10.3390/metabo16080553

Huang et al. 2026 Air Qual Atmos Health*PM_{2.5}-bound metals and female breast cancer incidence, New Mexico*

Census-tract analysis of New Mexico Tumor Registry breast-cancer incidence, 1997–2019, against annual PM_{2.5}-bound metal exposure intensities from a calibrated Emission Weighted Proximity Model over a **10-year pre-diagnosis window**, fitted by GEE with Gaussian estimation. All five metals were associated with higher age-adjusted incidence: **chromium RR 1.07 (1.05–1.08)**, **cobalt 1.08 (1.05–1.11)**, **copper 1.06 (1.04–1.07)**, **manganese 1.05 (1.02–1.08)**, **mercury 1.05 (1.03–1.07)**, with a monotone trend across unexposed/low/medium/high categories for every metal. The intervals are strikingly narrow, which is the tell: these are tract-level rate ratios on 23 years of registry data, so precision reflects the number of tract-years rather than the number of informative exposure contrasts, and the five metals are emitted by shared sources and are almost certainly collinear — five nominally independent estimates from one exposure surface. EWPM is a proximity model, not a measurement. Ecological throughout.

doi: 10.1007/s11869-026-02079-1

Mao et al. 2026 Front Public Health*Component-specific PM_{2.5} and endoscopically confirmed erosive gastric lesions*

Consecutive gastroscopies at Yixing People's Hospital, 1 Jan 2024 to 31 Dec 2025: **60,789 individuals, 14,325 with erosive gastric mucosal lesions**, against gridded daily PM_{2.5} and five chemical components in generalized additive models with attributable-burden analysis. Per IQR, **total PM_{2.5} IRR 1.076 (1.024–1.132)**; after FDR correction only **ammonium 1.082 (1.026–1.141)** and **sulfate 1.070 (1.024–1.118)** survived, with **nitrate, organic matter and black carbon all at 1.012** and intervals crossing the null. Endoscopic ascertainment removes the symptom-reporting bias that weakens most digestive air-pollution studies. The limitation is real and stated: the five components are strongly collinear and share one exposure model, so their independent contributions are not identified. Still the sharpest secondary-inorganic-versus-carbonaceous contrast available.

doi: 10.3389/fpubh.2026.1891236

Park et al. 2026b J Endocrinol Invest*Association between annual air pollutant exposure and thyroid function: a retrospective cross-sectional study*

16,193 adults (8,812 male, 7,381 female; mean age 53.8±13.1 y) screened at Seoul National University with TSH within 0.1–10.0 mIU/L, against CMAQ-estimated annual PM_{2.5}, PM₁₀, NO₂, SO₂ and CO. All five pollutants associate positively with TSH and negatively with free thyroxine ($P < 0.001$), with significant non-linearity for PM_{2.5} and NO₂ against log TSH. For the combined high-TSH/low-FT4 phenotype the per-IQR adjusted odds ratios run **PM₁₀ 1.55 (1.42–1.69)**, **SO₂ 1.38 (1.28–1.49)**, **NO₂ 1.36 (1.23–1.50)**, **PM_{2.5} 1.25 (1.17–1.34)** and **CO 1.23 (1.15–1.31)**, stronger among women. The binding limitations: exposure is a modelled annual field at residence with no personal component, the design is cross-sectional so temporality is unresolved, and the five pollutants are collinear by construction — the ordering must not be read as differential potency.

doi: 10.1007/s40618-026-02965-6

Wei et al. 2026 Environ Health (Wash)*Air pollution, lipid mediators and benign prostatic hyperplasia*

UK Biobank, **123,037 men**, with a western-China cohort replication. **2.18 % higher BPH risk per 1 µg/m³ PM_{2.5}** ($p = 0.0029$) in UK Biobank and **2.77 %** in the replication; mixture exposure by WQS and qgcomp gave 3.53 % and 2.20 % per unit for BPH and 6.78 % and 9.17 % for lower urinary tract symptoms. Restricted cubic splines put the **BPH risk threshold at 9.83 µg/m³**, and joint high-PM/high-fat-diet exposure carried 16 % higher risk than low/low. Thirty-four lipid metabolites are nominated as mediators. Two problems: BPH ascertainment in UK Biobank is by linked diagnosis and is sensitive to healthcare-contact patterns that correlate with urbanicity and therefore with PM_{2.5}; and 34 mediators from an untargeted panel without a stated multiplicity correction is hypothesis generation, not mediation analysis.

doi: 10.1021/envhealth.5c00806

Wu et al. 2026d J Am Coll Radiol*Short-term environmental exposures and emergency department imaging utilization*

Time-stratified case-crossover over **1,946,465 imaging studies** at five Toronto academic hospitals, 2013–2023, with conditional Poisson models against PM_{2.5}, NO₂, temperature, wind, rainfall and snowfall, modelling both absolute volume and per-visit rate (log daily visits as offset). Higher PM_{2.5} associates with higher imaging volume, **IRR 1.020 (95% CI 1.017–1.024) per 10 µg/m³**, alongside NO₂ 1.017 (1.014–1.020) per 10 ppb and temperature 1.040 per 10 °C; wind, rain and snow are protective, snowfall most strongly (0.959 per 10 mm). The volume effect is almost certainly mediated by visit counts rather than by per-visit imaging intensity, and the exposures are collinear with each other and with season — so this is a health-system demand signal, not evidence of a PM-specific clinical mechanism. Its value is operational: PM_{2.5} forecasts carry usable information for ED resource planning.

doi: 10.1016/j.jacr.2026.08.014

Zhang et al. 2026f Front Public Health*Plasma metabolomic signatures of air pollution and rheumatoid arthritis risk*

403,332 RA-free UK Biobank participants followed a median **14.01 years**, with residential annual PM_{2.5}, PM₁₀, NO₂, NO_x and a composite air pollution score, plasma metabolomics, machine-learning-derived pollutant-specific metabolic signatures, Cox models, generalised propensity score balancing and mediation analysis. Per SD of signature: **PM_{2.5}-related HR 1.208 (1.180–1.236)**, NO₂ 1.141, NO_x 1.174, APS 1.184, stable under propensity balancing. **Signatures accounted for 15.7–21.1 %** of the pollution–RA associations. The critical reading: the hazard ratios are for the *signature*, a metabolite score trained to predict exposure, not for exposure itself — a signature can outperform its own target because it absorbs correlates of exposure (diet, adiposity, smoking) that the pollutant variable does not. Single-timepoint metabolomics, UK Biobank's healthy-volunteer selection, and a mediation framework that assumes no exposure–mediator confounding.

doi: 10.3389/fpubh.2026.1901951

Zinan'dala et al. 2026 Int J Mol Sci*Urinary metabolomic alterations across contrasting ambient pollution regions, Thailand*

Untargeted ¹H-NMR urinary metabolomics comparing **51 adults in Chiang Mai (high exposure)** with **52 in Songkhla (low exposure)** during a regional pollution episode, using long-term residence as the exposure proxy. PLS-DA separated the groups ($R^2 = 0.873$ but $Q^2 = 0.447$ — good fit, modest prediction, and the gap is the standard warning sign of overfitting in this method). Eight metabolites differed at $p < 0.05$, six surviving Benjamini–Hochberg: **L-tryptophan, hippuric acid, xanthine and 5-HIAA remained significant after covariate adjustment**, with L-arginine and L-cystathionine lower in the high-exposure group. The design's fatal ambiguity is acknowledged by the authors: **region is the exposure variable**, and Chiang Mai and Songkhla differ in diet, water, altitude and season as much as in PM_{2.5}. Hippuric acid in particular is a dietary polyphenol marker before it is anything else. Tier B for method, not for attribution.

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Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Beydoun et al.	The interplay of air pollution with plasma neurodegenerative markers and metabolome for dementia, Parkinson's disease and all-cause mortality risks and transitions: the UK Biobank study	<i>Ecotoxical Environ Saf</i>	Cohort	Ambient PM _{2.5} / PM ₁₀ and NO ₂ /NO _x , modelled residential	United Kingdom
Chen et al.	Gender and season disparities in the influence of short-term PM _{2.5} exposure on emergency room visits for mood disorders in Taiwan	<i>J Occup Environ Med</i>	Time-stratified case-crossover + DLNM	Daily PM _{2.5} , lags 0-5	Taiwan
Hendrickx et al.	Effects of recent air pollution exposure on children's neurocognitive functions and the possible link with endothelial function	<i>Int J Environ Res Public Health</i>	Observational - cohort	Indoor and outdoor PM over 4 h, 24 h and 1 week windows; monthly NO ₂	Belgium
Ma et al.	Associations between short-term absolute increases in ambient air pollution and hospital admissions and outpatient visits for depression: A nationwide case-crossover study	<i>Ecotoxical Environ Saf</i>	Time-stratified case-crossover	Day-on-day absolute increase (APIN) in PM _{2.5} and PM ₁₀	China
Sakhvidi et al.	Ambient air pollution exposure and cognitive aging; mediators and possible pathways: a systematic review on the epidemiological studies	<i>Ageing Res Rev</i>	Systematic review	PM _{2.5} , PM ₁₀ , PM ₁ components, black carbon, NO ₂ , NO _x	Global
Wang et al.	Speciated fine particulate matter and self-reported sleep time in middle-aged and older Chinese adults	<i>Sleep Med</i>	Prospective cohort (CHARLS)	6-month and 2-year PM _{2.5} plus BC, OM, SO ₄ , NO ₃ , NH ₄ (TAP dataset)	China
Wu et al.	Early-life exposure to particulate matter and sleep in Chinese toddlers: the roles of particle size and particulate constituents	<i>Toxics</i>	Observational - cohort	Early-life PM ₁ , PM _{2.5} , PM ₁₀ and PM _{2.5} constituents from high-resolution grids	China
Cardiovascular & metabolic					
Chen et al.	Residential Incense Burning, Indoor Air Pollution, and Cardiovascular Hemodynamic Alterations	<i>Indoor Air</i>	Repeated-measures panel, 24 h indoor monitoring	24 h indoor PM _{2.5} and PM ₁₀ with CO, CO ₂ , TVOC, HCHO	Taiwan
Correa Lara et al.	Ambient particulate matter exposure and risk of venous thromboembolism: an updated systematic review and meta-analysis	<i>Thromb Haemost</i>	Review / synthesis	Long- and short-term PM _{2.5} and PM ₁₀ , harmonised to 10 µg/m ³	Global / multi-region
Li et al.	Sulfate-dominated PM _{2.5} constituents are associated with increased acute myocardial infarction mortality: a US nationwide difference-in-differences study	<i>Environ Sci Technol Lett</i>	Quasi-experimental (difference-in-differences)	PM _{2.5} mass and EC / OC / SO ₄ / NO ₃ constituent mixture, per IQR	United States
Ramirez-Moreno et al.	Regional disparities in stroke hospitalization and mortality associated with long-term fine particulate matter exposure: a nationwide ecological study in Spain	<i>Med Sci (Basel)</i>	Ecological	Regional annual mean PM _{2.5} (MITECO), region-year resolution	Spain
Yang et al.	Endothelial function in relation to low-level chronic residential air pollution in a general population: a cohort study	<i>Blood Press</i>	Observational - cohort	6-year residential PM ₁₀ 15.9, PM _{2.5} 12.8, NO ₂ 14.3, BC 1.04 µg/m ³ (spatiotemporal interpolation)	Belgium
Reproductive & developmental					
Chen et al.	Prenatal exposure to size-fractionated particulate matter and fetal growth trajectories: mediation by maternal glucose metabolism in a birth cohort	<i>Toxics</i>	Observational - cohort	Residential PM ₁ , PM _{2.5} , PM ₁₀ from 1 km spatiotemporal models	China
Kessler et al.	Joint Mixture Associations of Prenatal Air Pollution and Polyunsaturated Fatty Acid Intake on Birth Weight Outcomes in an African American Cohort	<i>Environ Health (Wash)</i>	Prospective birth cohort (mixtures)	First-trimester traffic-related PM _{2.5} , CO, NO _x	United States

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Wang et al.	Ambient Ultrafine Particles and Hypertensive Disorders of Pregnancy: A Mediation Analysis of Homocysteine in a Birth Cohort	<i>Environ Health (Wash)</i>	Prospective birth cohort	Modelled ultrafine particle number, high spatiotemporal resolution	China
Respiratory & allergic					
Aulia Anwar et al.	Smoking, environmental exposures, and chronic lung disease as key risk factors for lung cancer in North Sumatra	<i>Exp Oncol</i>	Case-control	Self-reported unhealthy indoor environment (ventilation, household pollutants)	Indonesia
China Lung Oncology Group	China Wumeng Mountain Lung Cancer: Evidence Chain, Clinical and Molecular Characteristics of Regional Lung Cancer Related to Coal Pollution in the Yunnan-Guizhou Plateau	<i>Zhongguo Fei Ai Za Zhi</i>	Multi-omics regional evidence synthesis	Household coal-combustion PM, source-resolved mineral composition	China
Li et al.	Associations Between Air Pollution and Hospitalization of Patients with Chronic Obstructive Pulmonary Disease in Kashi, China: A Time-Stratified Case-Crossover Study	<i>Int J Chron Obstruct Pulmon Dis</i>	Time-stratified case-crossover	Address-matched daily PM _{2.5} and PM ₁₀	China
Morvan et al.	When air pollution meets comorbidities: evidence on the causal effect of PM _{2.5} on COVID-19 hospitalizations and deaths	<i>Int J Epidemiol</i>	Spatial econometric model	Municipal weekly PM _{2.5} , instrumented by wind direction (2SLS)	France
Silva et al.	Shared mechanisms linking PM _{2.5} exposure, obstructive sleep apnea and obesity	<i>Front Sleep</i>	Mechanistic review	Chronic ambient PM _{2.5}	Global / multi-region
Wang & Li	Advances in research on the association between global air pollution and atopic dermatitis	<i>Front Public Health</i>	Narrative review	Ambient PM _{2.5} / PM ₁₀ and gaseous co-pollutants	Multi-region
Yang et al.	Analysis of Spatiotemporal Risks of Environmental Driving Factors of Respiratory Diseases	<i>GeoHealth</i>	DLNM + geographically-temporally weighted regression	Ambient PM and co-pollutants, prefecture daily	China
Zhongguo Fei Ai Za Zhi	China Wumeng Mountain lung cancer: evidence chain, clinical and molecular characteristics of regional lung cancer related to coal pollution in the Yunnan-Guizhou Plateau	<i>Zhongguo Fei Ai Za Zhi</i>	Narrative review	Household smoky-coal PM _{2.5} / black carbon, PAHs and methylated PAHs	China
de Souza et al.	Wildfire smoke, PM _{2.5} exposure and respiratory morbidity in a densely populated urban setting: spatial mapping and time-series evidence	<i>Sci Total Environ</i>	Time-series	Weekly mean CETESB PM _{2.5} ; INPE active-fire counts	Brazil
Mechanistic toxicology					
Chen et al.	Multi-omics integration reveals human placental molecular signatures associated with microplastics exposure	<i>Part Fibre Toxicol</i>	Human tissue multi-omics (transcriptome/proteome/metabolome)	Placental microplastic mass burden by Py-GC-MS, high vs low strata	China
Forest	Toxicological impact of sand and dust storms on human health: A review on the molecular mechanisms involved	<i>Sci Total Environ</i>	Mechanistic review	Storm-derived PM ₁₀ and PM _{2.5} (mineral, biological, metal fractions)	Multi-region
Li & Li	Research advances in the pulmonary carcinogenic effects of atmospheric fine particulate matter mediated by GPER	<i>Zhongguo Fei Ai Za Zhi</i>	Narrative review	PM _{2.5} estrogen-like organic components and metal ions	China
Li & Li	Research Advances in the Pulmonary Carcinogenic Effects of Atmospheric Fine Particulate Matter Mediated by GPER	<i>Zhongguo Fei Ai Za Zhi</i>	Mechanistic review	PM _{2.5} estrogen-like components and metal ions	China
Liu et al.	Ambient PM _{2.5} Induced Pulmonary Vascular Remodeling via MSC-Mediated Cellular Senescence in Pulmonary Arterial Endothelial Cells	<i>Environ Health (Wash)</i>	Real-ambient murine + cell model	Real-ambient PM _{2.5} exposure	China
Moonwiriyaikit et al.	Establishment of a STAT6 reporter assay for screening environmental toxicants affecting allergic airway inflammation	<i>J Vis Exp</i>	Toxicological (in vitro)	PM and B[b]F as test constituents; no ambient exposure metric	Not stated
Nikaido-Landry et al.	Epidermal growth factor receptor dysregulation and climate change: insights into pollution-promoted lung cancer	<i>Philos Trans R Soc Lond B Biol Sci</i>	Review / synthesis	Ambient fine PM as a promoter in EGFR-mutant histologically normal lung tissue	Global / multi-region

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Pathak et al.	Effect of airborne particulate matter on mtDNA copy number: A systematic review and meta-analysis	<i>medRxiv (preprint)</i>	Systematic review & meta-analysis	Ambient PM _{2.5} /PM ₁₀ , study-specific contrasts	Multi-region
Qin et al.	AlCl ₃ aerosol exposure is associated with pulmonary lactate accumulation, neutrophil dysfunction, and fibrosis-like remodeling	<i>Biofactors</i>	Animal + in vitro + human cohort	Repeated AlCl ₃ aerosol inhalation (dose not deposited)	China
Qin et al.	Engineered protein coronas reveal that apolipoprotein E enhances transport of particulate pollutants across the blood-brain barrier	<i>Environ Health (Wash)</i>	Toxicological (in vitro)	Aptamer-functionalised gold nanoparticles as PM surrogate; polystyrene and industrial-source PM	Not stated
TaRGET II Consortium	Lifecourse sex-specific molecular response to early-life exposures of toxic substances	<i>bioRxiv (preprint)</i>	Multi-omics animal consortium	PM _{2.5} among seven toxicants, preconception through lactation	United States
Van et al.	MASLD exacerbates chronic low-dose PM _{2.5} -induced lung injury, inflammation and fibrosis	<i>In Vivo</i>	Animal model (chronic instillation)	Diesel PM 0.5 mg/kg intratracheal 3x/week (50 µg/m ³ equivalent)	Taiwan
Occupational & indoor					
Li & Zheng	Size-resolved PM ₁₀ -bound PFASs: respiratory deposition and association with oxidative stress biomarkers among industrial waste recycling workers	<i>Toxics</i>	Observational - cohort	Size-resolved PM-bound PFAS (PM _{2.1} , PM _{2.1-10} and cut-point bins); In-ADI	China
Ospalek et al.	Air quality assessment and potential inhalation exposure across equestrian facilities: a case study	<i>Atmosphere</i>	Cross-sectional environmental survey	PM ₁ , PM _{2.5} , PM ₁₀ , respirable and total dust at 1.5 m; NH ₃	Slovakia
Perna et al.	Impact of indoor air pollution on occupational exposure: rethinking the major health threat for airport workers	<i>Life (Basel)</i>	Review / synthesis	Ultrafine PM, black carbon, PAHs and trace metals in airport indoor microenvironments	Global / multi-region
Ramirez et al.	Airway diseases and occupational exposures in poultry industry workers: a comprehensive review	<i>Curr Allergy Asthma Rep</i>	Comprehensive review	Poultry-barn organic dust: feather, mite, feed dust, mould, endotoxin, NH ₃	Global / multi-region
Sadki et al.	Real-time monitoring of formaldehyde exposure and acute health symptoms in healthcare workers in Morocco: a cross-sectional study	<i>Cad Saude Publica</i>	Cross-sectional survey	Shift-long airborne formaldehyde by calibrated semiconductor dosimeter (gas, not PM)	Morocco
Siemens et al.	A SynAir-G position statement for the improvement of indoor air quality for children	<i>Indoor Air</i>	Consensus position statement	Indoor chemical and biological pollutants, bioaerosols, synergistic mixtures	Europe
Yang et al.	Bioaerosols in the atmospheric environment of oil exploitation sites: spatial and temporal distribution, sources, influencing factors, health risks, and reduction efficiency of masks	<i>Environ Pollut</i>	Seasonal field sampling + filtration test	Culturable and total bioaerosol, particle-bound microorganisms	China
Sensing, forecasting & instrumentation					
Alharbi et al.	Multi-Technique Characterization of Atmospheric Aerosol Particles from the Coastal Area of Jeddah, Saudi Arabia: Morphology, Surface Chemistry, and Mineralogy	<i>Atmosphere</i>	Field campaign	PM _{2.5} morphology, surface oxidation state, crystalline phase (SEM/EDS/XPS/XRD)	Saudi Arabia
Anon et al.	Assessment Campaign of the Gravimetric Method (PM ₁₀ and PM _{2.5}): Analyzing Environmental Artifacts and Metrological Consistency for the 2030 Regulatory Challenge	<i>Atmosphere</i>	Interlaboratory comparison	Gravimetric PM ₁₀ and PM _{2.5} (EN 12341)	Italy
Chen et al.	Study on the pollution characteristics of aerosols and VOCs along the lower reaches of the Yangtze River based on shipborne observations	<i>Atmos Environ</i>	Metadata only (no abstract)	Shipborne aerosol and VOC concentrations (not extractable)	China
Eger et al.	Applicability of Compact Low- and Mid-Cost Sensors to Monitor Air Pollutant Emissions from Inland Ships	<i>Aerosol Air Qual Res</i>	Year-long co-location field evaluation	BC, ultrafine particle number, NO/NO ₂ , CO ₂ in ship plumes	Germany

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Fahim & Karim	Cost-Aware Evaluation of Time-Series Foundation Models for Urban Air-Quality and Temperature Forecasting	<i>Res Sq (preprint)</i>	Benchmark / forecasting model comparison	Hourly PM _{2.5} concentration	Multi-country (29 cities)
Giannetti et al.	Ants as bioindicators of urban traffic pollution	<i>Environ Pollut</i>	Field study	Magnetic susceptibility of traffic-derived PM on ant exoskeleton	Parma, Italy
Kang et al.	Emission characteristics and human health impacts of vehicle originated condensable particulate matter	<i>J Hazard Mater</i>	Multi-instrument field evaluation	Real-time condensable PM mass and number (PNU-CPM system); CPM emission factors	South Korea
Lefevre et al.	Urban microclimate dataset from low-cost sensor networks in a hot and humid tropical climate: Reunion Island	<i>Data Brief</i>	Network deployment + field evaluation	None; T, RH, wind, GHI, grey-globe temperature at 10-min cadence	Reunion Island (France)
Li et al.	Early warning of cucumber angular leaf spot by estimating airborne pathogen aerosols with particulate matter sensors	<i>Plants (Basel)</i>	Network deployment + field evaluation	Low-cost PM sensor concentration as a physical proxy for pathogen aerosol load	China
Marcon et al.	Citizen science in air pollution monitoring: methodological approaches, effectiveness, and integration with health research	<i>Curr Environ Health Rep</i>	Critical review	Low-cost sensor networks, wearables, AI-assisted calibration	Global / multi-region
Moteki et al.	Constraining the refractive index of fractal-like aggregates from complex forward-scattering amplitudes of mass-classified aerosol particles: application to black carbon samples	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Complex refractive index of mass-classified black carbon aggregates	Not stated
Neuberger et al.	Importance of hydrated aerosol particles for aerosol-fog relationships in the Italian Po Valley	<i>Atmos Chem Phys</i>	Field campaign + LES modelling	Hygroscopicity parameter kappa; hydrated vs activated particle diameter	Italy
Roska et al.	Beyond targeted calibrations: comprehensive quantification of oxygenated organic compounds in urban atmospheres	<i>Environ Sci Technol</i>	Instrument development	OVOC precursors of SOA; NH ₄ ⁺ -adduct CIMS	United States + Germany
Vivanco et al.	Assessing the responsiveness of an air quality model to real-world emission changes for air quality planning	<i>Environ Pollut</i>	CTM evaluation against observation-based counterfactuals (COVID-19 emission shift)	Modelled NO ₂ and O ₃ response to a real-world emission reduction; PM not reported in abstract	Europe
Wang et al.	Perchloric acid (HClO ₄) drives atmospheric new particle formation enhanced by dimethylamine, ammonia and sulfuric acid	<i>Atmos Chem Phys</i>	Kinetic model + data reanalysis	Cluster formation rate for PA-DMA / PA-NH ₃ / PA-SA nucleation	Arctic / coastal marine
Werderman et al.	Laboratory evaluation of Sensirion SEN66, Piera IPS-7100, PurpleAir PIXEL and OPC-N3 low-cost aerosol sensors across aerosol types and concentration ranges	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Chamber PM across aerosol types (not extractable)	Not stated
Wongsasuluk et al.	PM _{2.5} exposure and respiratory health risk for vulnerable populations in an urban underpass: a predictive MLR approach, Chiang Mai	<i>Environ Sci Pollut Res Int</i>	Network deployment + statistical model	PM _{2.5} and PM ₁₀ from five DustBoy low-cost stations, 10-min cadence	Thailand
Xiong et al.	In situ real-time determination of SO ₂ photochemical oxidation in nanoscale sea salt aerosols based on dark-field microscopy	<i>Atmos Chem Phys</i>	Laboratory instrument development	Single-nanoparticle hygroscopic growth factor; first-order rate constant	China
Yu et al.	Machine learning-based prediction of occupant symptom risk using indoor and outdoor environmental quality data in university dormitories	<i>Indoor Air</i>	Continuous IEQ monitoring + supervised classification	Five indoor and five outdoor environmental parameters incl. PM; distributional shape features	Italy
Zhang et al.	Buffering of atmospheric nanoparticle growth by temperature-dependent shifts in molecular composition, volatility and diffusivity	<i>Atmos Chem Phys</i>	Kinetic model + data reanalysis	Nanoparticle growth rate (nm/h) vs condensable organic vapour	Global / multi-region

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Zhang et al.	Contrasting nighttime heterogeneous and daytime photochemical aging drive the optical evolution of black carbon	<i>Atmos Chem Phys</i>	Field campaign, single-particle instrumentation	rBC mass, core MMD, coating-to-core mass ratio, MAC at 532 nm (SP2 + SPAMS)	China
Exposure assessment & modelling					
Agpaoa et al.	Occurrence, temporal variation and meteorological drivers of airborne fungal spores in Laoag City, northern Philippines	<i>Environ Monit Assess</i>	Bioaerosol monitoring campaign	Volumetric spore counts (spores/m ³), six spore types	Philippines
Anon et al.	Severe winter PM _{2.5} pollution in remote Canadian mountain communities: Insights into the role of condensable particulate matter under subzero conditions	<i>Environ Int</i>	Field campaign	PM _{2.5} including condensable fraction, subzero conditions	Canada
Anon et al.	Metal Processing and Combustion Dominate Source-Specific PM _{2.5} Oxidative Potential in a National Shipbuilding Complex	<i>Atmos Pollut Res</i>	Field campaign	Source-apportioned PM _{2.5} oxidative potential	South Korea
Chen et al.	Source-specific oxidative potential of PM _{2.5} in Xi'an: roles of water-soluble metals revealed by DTT assay and interpretable machine learning	<i>Toxics</i>	Source apportionment	Volume-normalised DTT oxidative potential (DTTv) of ambient PM _{2.5}	China
Cui et al.	Daily 1 km satellite-based PM _{2.5} exposure estimates in Bangladesh, 2020-2025	<i>Environ Pollut</i>	Satellite-surface fusion, machine learning	Gap-filled MA-IAC AOD -> ground PM _{2.5} (LightGBM + TabNet)	Bangladesh
Hopke et al.	A PM _{2.5} magnitude-rank index for summarizing annual concentration-frequency structure	<i>Environ Monit Assess</i>	Metric development + retrospective application	PMRI: largest k with >= k days at >= k µg/m ³	United States
Isworo & Oetari	A risk-integrated framework for industrial air quality assessment linking dispersion modeling and health-based environmental pressure indicators	<i>Environ Sci Pollut Res Int</i>	Modelling / inventory	Gaussian-plume PM _{2.5} and PM ₁₀ ; hazard-quotient-weighted AEPI	Indonesia
Kalumbi et al.	Age-specific time-spent patterns among residents living near gold mine tailings storage facilities in the West Rand, South Africa	<i>J Expo Sci Environ Epidemiol</i>	Cross-sectional time-activity survey	None measured; indoor/outdoor time fractions as exposure-weighting inputs	South Africa
Kang et al.	Characteristics and dry deposition fluxes of particulate parent, alkyl, and nitro-polycyclic aromatic hydrocarbons in the Bohai Sea Strait, China	<i>Atmos Environ</i>	Metadata only (no abstract)	Particle-bound parent / alkyl / nitro-PAH (not extractable)	China
Kim et al.	Application of machine learning-based wind corrections to coupled WRF-CMAQ simulations: a comparison of dynamic assimilation and static replacement	<i>Atmos Environ</i>	Metadata only (no abstract)	Not extractable	Not stated
Leibensperger et al.	Sea surface warming suppresses primary sea spray aerosol number production	<i>Atmos Chem Phys</i>	Controlled wave-channel experiment	Bubble and SSA number concentration and emission flux, 0.008-20 µm	Global / multi-region
Li et al.	Study on the transport mechanism of surface particulate matter based on the closed earth-air model	<i>Environ Pollut</i>	Measurement campaign	PM ₁ , PM _{2.5} and PM ₁₀ in a semi-enclosed basement; earth-air flow, CO ₂ , 222Rn	China
Mainka & Mainka	Urban cycling vs. PM _{2.5} exposure: does the benefit outweigh the risk?	<i>Preprints.org</i>	Proxy validation vs personal exposure	Trip-resolved personal PM _{2.5} from low-cost sensors; inhaled dose via ventilation rate	Poland
Mandalapu et al.	Joint Heat and PM _{2.5} Exposure Across US Metropolitan Areas: Multi-Stressor Disparities, Historical Redlining, and a Multi-Metric Assessment Framework	<i>medRxiv (preprint)</i>	Ecological cross-sectional, copula tail dependence	Summer mean PM _{2.5} , tract-level, jointly with surface heat	United States
Mohammed	Spatiotemporal dynamics of atmospheric vulnerability in hot arid cities: introducing the Urban Microclimate Air Stress Index	<i>Build Environ</i>	Metadata only (no abstract)	Composite microclimate/air-stress index (not extractable)	Hot arid cities (not stated)
Moryani et al.	Inhalation bioaccessibility, pollution assessment, and speciation of heavy metals in PM _{2.5} road dust from urban and semi-urban areas of Pakistan	<i>Toxics</i>	Measurement campaign	PM _{2.5} -fractionated road dust; ALF-bioaccessible Cr, As, Pb	Pakistan

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Mueangsai et al.	Factors Associated with Health Risk Quotient among Communities Exposed to PM _{2.5} from Stone Crushing Plants in Chonburi Province Thailand	<i>Res Sq (preprint)</i>	Community cross-sectional + USEPA HQ	Fixed-site PM _{2.5} , two stations, community average	Thailand
Rashid et al.	Ultrafine particles and mortality: a scoping review of epidemiologic evidence and exposure assessment limitations	<i>Toxics</i>	Review / synthesis	Ambient ultra-fine particle number concentration (<100 nm)	Global / multi-region
Reyna-Bello et al.	Seasonal variability of PM _{2.5} -bound metals and meteorology in an urban-industrial area of northern Mexico: a PCA assessment	<i>Environ Monit Assess</i>	Filter sampling + composition	MiniVol low-volume PM _{2.5} mass; 10 metals by FAAS	Mexico
Walker et al.	Divergent social patterning of directly measured environmental exposures across Rhode Island communities	<i>medRxiv (preprint)</i>	Direct-measurement field campaign	Short-term PM _{2.5} alongside LAeq/LCeq noise and night-time illuminance	United States
Wu et al.	Global fire activity exacerbates extreme PM _{2.5} pollution	<i>npj Clim Atmos Sci</i>	Metadata only (no abstract)	Extreme-percentile PM _{2.5} attributable to fire activity (not extractable)	Global / multi-region
Zhang et al.	Photoinduced intraparticle chemistry drives the rapid atmospheric oxidation of biomass burning organic aerosol	<i>Environ Sci Technol</i>	Chamber + aircraft	Biomass-burning organic aerosol oxidation state	United States
Zhang et al.	Improved representation of isoprene-derived secondary organic aerosol in CAM6-Chem reveals regional contrasts in its long-term changes over China	<i>Atmos Chem Phys</i>	Chemical transport modelling	Isoprene-derived SOA mass and subspecies composition	China
Zhong et al.	Photochemical ultrafine particles pose potential inflammatory risk despite low PM _{2.5} mass	<i>Environ Int</i>	Observational + ML	Lung-deposited surface area vs PM _{2.5} mass	Fuzhou, China
Burden, policy & mitigation					
Acat et al.	National trends in ambient air pollutants and cause-specific mortality in Türkiye, 2018-2024: an ecological time-trend study	<i>Atmosphere</i>	Monitoring-record analysis	Unweighted national mean station PM ₁₀ (50.4 -> 41.0 µg/m ³); PM _{2.5} excluded	Türkiye
Anon et al.	Lifetime risk of respiratory system cancers across 185 countries: Global variation, socioeconomic inequalities, and ecological associations with air pollution	<i>Prev Med Rep</i>	Ecological (GLOBOCAN secondary analysis)	Country-level annual PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂	Multi-region
Anon et al.	A Harmonization Index for Comparative Assessment of Industrial Particulate Matter Emission Regulations	<i>Res Sq (preprint)</i>	Regulatory coding framework	Industrial PM emission limits across regulatory architectures	Multi-region
Cliff et al.	How do climatic factors directly influence the incidence and risk of meningococcal meningitis across the African meningitis belt? A narrative literature review	<i>GeoHealth</i>	Narrative review	Atmospheric mineral dust load and wind speed	Sub-Saharan Africa
Nilsson et al.	Urban environment, health and equity: symptoms and annoyances related to air pollution, traffic noise and greenness	<i>Environ Epidemiol</i>	Cross-sectional population survey	Dichotomised residential PM _{2.5} , PM ₁₀ , NO ₂ at geocoded address	Sweden
Ozmen et al.	Long-term ambient PM _{2.5} exposure and premature mortality across Türkiye	<i>Toxics</i>	Burden estimation (GBD)	Province-level annual mean PM _{2.5} , counterfactual 5 µg/m ³ (WHO AirQ+)	Türkiye
Sharma et al.	Geospatial linkage of environmental exposures, socioeconomic profile and health outcomes among women of reproductive age, Victoria	<i>Health Promot J Austr</i>	Ecological geospatial linkage	Postcode-level PM _{2.5} , NO ₂ , CO, SO ₂ , HCHO, O ₃ , LST, greenness	Australia
Sirikul et al.	Knowledge, attitudes and practices of caregivers regarding PM _{2.5} exposure among children in northern Thailand	<i>J Clim Chang Health</i>	Cross-sectional survey	None measured; AQI literacy and protective behaviour	Thailand
Sommar et al.	Health economic assessment of PM _{2.5} originating from residential wood combustion - a case study in northern Sweden	<i>Int J Environ Res Public Health</i>	Burden estimation (GBD)	Population-weighted RWC-attributable PM _{2.5} , 0.16 µg/m ³ (IQR 0.10-0.21)	Sweden

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Wang et al.	Trends, inequalities, and 2030 projections of women's tracheal, bronchial, and lung cancer attributable to air pollution, radon, and tobacco exposure	<i>Front Glob Womens Health</i>	Burden estimation	Ambient particulate matter and household solid fuel (GBD risk factors)	Global
Zhai et al.	Global disparities in lifetime risk of lung cancer: links to socioeconomic development, smoking, and fine particulate matter from GLOBOCAN 2022	<i>Front Public Health</i>	Ecological	Country-level PM ₁ and PM _{2.5} from global environmental databases	Global / multi-region
Zhang et al.	Health and economic impacts of improved air quality from transitioning global shipping energy to hydrogen	<i>Nat Commun</i>	Integrated emission-air quality-health modelling	Modelled PM _{2.5} and O ₃ concentration change under hydrogen scenarios	Global / multi-region
Zhu et al.	Short-term effectiveness of urban road-cleaning operations for fugitive road dust control: evidence from TRAKER-based mobile monitoring	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	TRAKER-derived road-dust emission potential (not extractable)	Not stated
Other clinical endpoints					
Chen et al.	Untargeted metabolomics reveals metabolic perturbations in community-dwelling elderly exposed to PM _{2.5} -bound ester compounds	<i>Metabolites</i>	Observational - cross-sectional	72-h personal PM _{2.5} (UPAS V2), mean 38.06 $\mu\text{g}/\text{m}^3$; 20 GC-MS-prioritised esters	China
Huang et al.	Long-term exposure to PM _{2.5} -bound metals and female breast cancer incidence in New Mexico, USA	<i>Air Qual Atmos Health</i>	Ecological	Census-tract PM _{2.5} -bound Cr, Co, Cu, Mn, Hg over a 10-year pre-diagnosis window (EWPM)	United States
Mao et al.	Component-specific effects of PM _{2.5} on endoscopically identified erosive gastric mucosal lesions: a time-series study in Yixing, China	<i>Front Public Health</i>	Time-series (generalized additive model)	Gridded daily PM _{2.5} and five chemical components	China
Park et al.	Association between annual air pollutant exposure and thyroid function: a retrospective cross-sectional study	<i>J Endocrinol Invest</i>	Retrospective cross-sectional (health screening)	Annual mean PM _{2.5} and PM ₁₀ from CMAQ, per IQR	South Korea
Wei et al.	Prospective Metabolomic Cohort of 123,037 Men Reveals Lipid Mediators Linking Air Pollution to Benign Prostatic Hyperplasia	<i>Environ Health (Wash)</i>	Prospective cohort (mixtures)	Annual PM _{2.5} , PM ₁₀ , NO ₂ , NOx; WQS and qgcomp mixtures	United Kingdom
Wu et al.	Short-term environmental exposures and emergency department imaging utilization	<i>J Am Coll Radiol</i>	Time-stratified case-crossover	Daily PM _{2.5} , per 10 $\mu\text{g}/\text{m}^3$	Canada
Zhang et al.	Plasma metabolomic signatures of air pollution and rheumatoid arthritis risk	<i>Front Public Health</i>	Observational - cohort	Residential annual PM _{2.5} , PM ₁₀ , NO ₂ , NOx and a composite air pollution score	United Kingdom
Zinan'dala et al.	Urinary metabolomic alterations associated with contrasting ambient air pollution exposure in Thai adults: an untargeted 1H-NMR study	<i>Int J Mol Sci</i>	Observational - cross-sectional	Regional residence as proxy for PM _{2.5} contrast (Chiang Mai vs Songkhla)	Thailand

Provenance and method

Period 2026-08-23 to 2026-08-29, seven daily issues pooled. **This rollup queried no API.** It aggregates `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over `PMRW_START-PMRW_END`; every record was screened, summarised and DOI-gated at daily-issue time and is reproduced verbatim. **109 records, 107 unique DOIs** — the two duplicates are named on the venues page. Daily totals: 9, 9, 14, 14, 36, 19, 8. **Week-over-week comparisons** are against the 16–22 August rollup (112 records) and, where a longer run is cited, the 8 and 15 August rollups. **Effect estimates** are pooled as recorded, on heterogeneous contrasts, with no meta-analysis. **Figures** are regenerated over the pooled range, not composited from the daily images. **Labels** — subtopic, design, particle-metric, geography — are single-label by dominant contribution, assigned at daily-issue time, and are a judgement rather than a controlled vocabulary.